

Pop-stacked permutations (OEIS A307030 and A309993)

The explicit generating function and its poles;
an operator proof of the leading asymptotic, with inversion;
the run distribution and the fixed-run denominators

Three research reports of one external research session
(bundle Reports 164, 161 and 166), merged into one ProveIt report

All three Parts: 3 October 2026 merged: 6 October 2026

Abstract

Let p_n be the number of permutations of $[n]$ in the image of one pass of pop-stack sorting (OEIS A307030), with $p_0 = 1$, and let $p_{n,k}$ count those with k maximal ascending runs (OEIS A309993). This report prints three manuscripts on these numbers. Part I gives an explicit exponential generating function $P = N/D$, a quotient of entire functions with no square-root branch; it proves that every pole is simple, with an explicit residue, that there are infinitely many positive poles, one for each solution of an explicit phase equation, and hence that P is not D-finite and (p_n) is not P-recursive; it shows that P is nevertheless differentially algebraic, gives a Riccati equation that computes $p_0, \dots, p_N$ in $O(N^2)$ arithmetic operations, isolates the dominant pole ρ by a positive-operator argument, gives the amplitude C in closed form and certifies both to about fifty digits. These are the numerical predictions of Claesson, Guðmundsson and Pantone (2019), now proved. Part II proves the leading asymptotic $p_n = C n! \rho^{-n} (1 + O(q^n))$ by a self-contained spectral and analytic Fredholm argument, without the explicit formula, with an elementary bracket for ρ and an all-orders expansion of the continuous inverse. Part III gives the bivariate generating function by runs, the mean and variance of the run count with exponentially small errors, a central limit theorem and a uniform lattice local limit theorem, and an alternate proof of the fixed-run denominator conjecture (Conjecture 2) of Claesson, Guðmundsson and Pantone; a proof posted by Shivam Patel on 20 August 2026 came first, and Part III credits it. The write records, with an interval-arithmetic proof, that the list of positive poles printed by Claesson, Guðmundsson and Pantone omits the thirteenth pole $\rho_{12} = 5.33887671271979\dots$ (their repository lists it), confirms that each of their fifteen printed positive values is a correct truncation, places the inverses relative to the repository's transseries volume, and checks Patel's argument. Effective error constants and onsets, the ordering of the nonreal poles, a minimal differential equation, bit complexity and several finer probabilistic statements remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 602e5bd0f (batch 108, cluster 108-POP). All three are dated 3 October 2026, and all three study pop-stacked permutations: the permutations in the image of one pass of pop-stack sorting, which reverses every maximal decreasing run. By the characterization of Asinowski, Banderier, Billey, Hackl and Linusson [1, 2], these are the permutations whose consecutive maximal ascending runs, with endpoints (a, b) and (c, d) , satisfy $a < d$ and $c < b$.

Part	Bundle report	Delivered title	Labels
I	164 (base)	Exact generating function for pop stacked permutations: Poles, differential algebraicity and faster enumeration	psp:egf:
II	161	Factorial exponential growth of pop stacked permutations: An operator proof and asymptotic inversion	psp:op:
III	166	Run distribution in pop stacked permutations: An explicit bivariate generating function and Gaussian laws	psp:run:

One model. All three Parts start from the same integral model: a run is a *state*, either a singleton $x \in (0, 1)$ or an interval (a, b) with $0 < a < b < 1$; consecutive runs are joined by the overlap kernel $\mathbf{1}\{\text{lo}(s) < \text{hi}(t), \text{lo}(t) < \text{hi}(s)\}$; a singleton has weight z and an interval $z^2 e^{z(b-a)}$; and the exponential generating function is 1 plus a bilinear pairing with the resolvent of the resulting compact operator (Part I, (2.4); Part II, (15.2); Part III, (22.4), where every run also carries a factor u). Each Part proves this normalization itself. Parts I and III then reduce the resolvent to a two-point boundary problem and solve it in closed form; Part II does not solve it, and argues spectrally throughout. All three isolate the dominant pole by the same Perron argument and the same strict comparison on the circle $|z| = \rho$.

Arrangement. The base of the merge is Report 164; its `Report164.tex` was staged as this `article.tex`, and it is printed first, as Part I. It is the fullest statement on A307030 itself: it contains Report 161’s main theorem as its Theorem 9.2, with the amplitude in closed form, and it has the pole set, the residues, non-D-finiteness, differential algebraicity, the Riccati recurrence and the certificate, none of which the other two prove. Report 161 is printed second, as Part II: its operator proof is a second route to Part I’s leading asymptotic, with the Fredholm continuation written out, and its all-orders inverse goes beyond Part I’s first Newton step. Report 166 is printed third, as Part III: its bivariate formula reduces to Part I’s at $u = 1$ (a second derivation of Part I’s formula), and its run statistics and fixed-run theorem are new relative to the other two. The order follows the univariate theory first; the manuscripts were written in the order 161, 164, 166 within about an hour and a half, and 164 and 166 each say that the earlier reports are “unchanged”, but none cites another in its bibliography or uses another’s results (Part I’s count fixture is derived from Report 161’s count file; see the Provenance).

Numbering. Sections are numbered continuously, and statements and equations within sections, as in all three manuscripts. Part I keeps Report 164’s Sections 1–13 and all its numbers. Report 161’s Section k is Section $k + 13$ here (Sections 14–20), and Report 166’s Section k is Section $k + 20$ (Sections 21–32); their statements and equations move with their sections (Report 161’s Theorem 1.1 is Theorem 14.1, Report 166’s Theorem 10.1 is Theorem 30.1; the README maps them all).

Section 33 and Remark 5.1 were added in the write; no delivered number of Part I moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (6 October 2026). They give cross-references between the Parts, record what the write checked, and record what later Parts and the write’s own computations say about earlier text. No delivered statement was changed and no delivered symbol was renamed. Section 33 gathers the open questions of all three Parts under Vladimir’s standing rule of 4 October 2026, with their sources, sketches and what is missing.

What is proved, and where

Statement	Where
<i>The total count A307030</i>	
Explicit EGF $P = N/D$ with entire N , D and no branch; local trigonometric form	Part I, Theorem 1.1, (1.5), (1.6); second derivation: Part III at $u = 1$
Pole set $\{D = 0, Q \neq 0\}$; every pole simple, residue $-2e^{-2\zeta}(4e^\zeta - 1)/(\zeta + 2)$; zeros of Q removable; finite-radius pole expansions	Part I, (1.7), (1.8), Section 6, (6.3)
Exactly one positive pole ρ_k for each $k \geq 0$; P not D-finite, (p_n) not P-recursive; the ordinary series not D-finite	Part I, Section 5
The positive-pole list printed by Claesson–Guðmundsson–Pantone omits ρ_{12} ; its fifteen values are correct truncations	[write] Remark 5.1 (interval proof)
Differentially algebraic of order at most two (resultant annihilator)	Part I, (7.1)
Riccati equation; $p_0, \dots, p_N$ in $O(N^2)$ arithmetic operations and $O(N)$ storage	Part I, (6.1), (8.1)
Unique dominant pole ρ ; $p_n = C n! \rho^{-n} (1 + O(\theta^n))$, $C = 2e^{-2\rho}(4e^\rho - 1)/(\rho(\rho + 2))$	Part I, Theorem 9.2; second route: Part II, Theorem 14.1 (C in spectral form)
Rational intervals of width 10^{-56} for ρ and 10^{-48} for C	Part I, (10.1); Part III, (28.1) (same L , U)
$(24/23)^{1/4} \leq \rho \leq 2$ by elementary counting	Part II, (14.7)
Fixed-order Stirling expansions; two-ceiling bracket for the threshold $N(y)$; one Newton step (Part I) and every fixed order (Part II) of the continuous inverse	Part I, Section 11; Part II, Corollary 19.1, Theorem 19.2, Proposition 19.3
<i>Runs, A309993</i>	
Bivariate EGF $P(z, u) = Q(H + BJ_0)/(h_0(BH - QJ_0))$	Part III, Theorem 21.1
$\mathbb{E}K_n = \mu n + \beta + O(\eta^n)$, $\text{Var } K_n = \nu n + \gamma + O(\eta^n)$, closed forms for μ, ν with rational certificates; CLT; uniform lattice local limit with error $O(n^{-1})$	Part III, Theorem 21.2, Sections 27–29
Fixed-run denominators $\prod_{j \leq k} (1 - jx)^{k-j+1}$, numerator degree $k(k+1)/2$, leading coefficient $(-1)^{k(k+1)/2+1} \prod_{j \leq k} j!$ (Conjecture 2 of Claesson–Guðmundsson–Pantone; first proved by Patel)	Part III, Theorem 30.1 (an alternate proof); the write’s check of Patel’s proof in the front matter (next section)
<i>Open</i>	
Effective complex gap, error constants and onsets (all Parts); ordering of the nonreal poles and convergent pole sums; a minimal-order differential equation; bit complexity and optimal cost of counting; the global bivariate polar set, minimal fixed-run denominators, Edgeworth and large-deviation refinements, certified β, γ . See Section 33.	

The finite computations of the three packages (exact integers and rationals, rational interval certificates) check identities, coefficients and transcriptions and certify the displayed constants; no asymptotic statement rests on them.

The predictions of Claesson, Guðmundsson and Pantone, and their status

Claesson, Guðmundsson and Pantone [3] (arXiv:1908.08910, version 1 of 23 August 2019; the write read it and the 12-page author PDF, whose pagination and numbers are the same) computed p_n for $n \leq 1000$ and analysed the sequence experimentally. They wrote a_n for p_n and μ for the dominant singularity. Their statements, and where this report stands on each:

- *An open enumeration problem* (Section 1, p. 2): “finding a generating function or a closed form solution to their enumeration remains an open problem.” Part I’s Theorem 1.1 gives the exponential generating function in closed form.
- *Counting cost* (Section 2, p. 5): their algorithm computes all counts up to N “in $O(N^4)$ time using $O(N^3)$ memory”, and they add that this “assumes that all arithmetic operations are $O(1)$ ”. Part I’s recurrence (8.1) uses $O(N^2)$ arithmetic operations and $O(N)$ stored numbers in the same model; Part I claims no bit complexity and no optimality.
- *Automated fitting* (Section 3.1, pp. 7–8): from 1000 terms they found no rational, algebraic, D-finite or D-algebraic fit for the ordinary generating function or for transforms of it including the exponential generating function, and derived rigorous size bounds for the rational, algebraic and D-finite cases (a D-finite equation would need $(k+2)(d+1) > 1000$). Part I’s non-D-finiteness makes the D-finite bound moot, and its D-algebraic annihilator (7.1) is consistent with their bounded search (Part I says so; the write did not compute where the annihilator falls in their search space).
- *Conjecture 2* (Section 3.2, p. 9): for every k the ordinary generating function of the permutations with k runs is $F_k(x) = N_k(x) / \prod_{i=1}^k (1 - ix)^{k-i+1}$, “where $N_k(x)$ is a polynomial of degree $k(k+1)/2$, the same degree as the conjectured denominator.” **Proved:** first by Shivam Patel (below), then by Part III’s Theorem 30.1, an alternate proof that also gives the leading coefficient.
- *Positive real singularities* (Section 3.3, p. 10): fifteen values, the first nine to 75 decimals and the rest to 56, 47, 36, 18, 11 and 6 decimals, the last two 5.55979961612 and 5.659669. “Each of these singularities is predicted to have critical exponent -1 , making them simple poles.” **Proved**, and more: every pole of P is simple (Part I, Section 6), and the positive poles are exactly the solutions $\rho_0 < \rho_1 < \dots$ of the phase equation (5.4). The write’s interval computation shows that each printed value is a correct truncation of some ρ_k , and that the list is $\rho_0, \dots, \rho_{11}, \rho_{13}, \rho_{14}, \rho_{15}$: it omits $\rho_{12} = 5.33887671271979\dots$ (Remark 5.1). The authors’ repository file [12], added three days before the arXiv posting, lists ρ_{12} ; the omission is in the printed list only.
- *Infinitely many singularities and non-D-finiteness* (Section 3.3, p. 10): “These results suggest that the exponential generating function may possess an infinite number of singularities. If true, this would imply the non-D-finiteness of both the ordinary and exponential generating functions.” **Proved** (Part I, Section 5, including the ordinary series).

- *Complex pairs* (Section 3.3, p. 10): three pairs, “also simple poles”, near $0.42794 \pm 3.60126i$, $1.80793 \pm 4.04623i$, $2.50840 \pm 4.24168i$, printed to 73 decimals. Every pole is simple (Part I). Newton’s method on Part I’s D started from the printed digits converges, in 220-digit arithmetic, to zeros of D with $Q \neq 0$, hence (by Part I’s pole-set theorem) to poles. The printed digits are correct to 62–66 decimal places, after which each printed string drops two consecutive digits; the repository’s values agree with these zeros to at least 108 decimal places (Remark 5.1). This is a numerical observation, not an interval certificate, and Part I proves nothing about the positions of nonreal poles beyond the equation $D = 0$.
- *The constants* (Section 3.3, pp. 10–11): $\mu \approx 1.11343904173672704376 \dots$ (75 decimals printed), $\mu^{-1} \approx 0.89811831857468696951 \dots$ (34 decimals printed) and $C \approx 0.69568854907063576799 \dots$ (58 decimals printed), with $a_n \sim C \cdot n! \cdot (0.898118 \dots)^n$. **Proved** with an exponentially small relative error (Part I, Theorem 9.2; Part II, Theorem 14.1); C has the closed form (9.2). Part I’s certificate (10.1) proves 56 decimals of ρ and 48 of C ; the write’s interval check proves all 75 printed decimals of ρ , and its 220-digit evaluation of the closed form agrees with all 58 printed decimals of C and all 34 of μ^{-1} (not interval-certified beyond Part I’s 48 decimals for C).

Asinowski, Banderier and Hackl [2] (DMTCS 22:2, 2021; read by the write) report these predictions in their Section 2.4 (p. 18), where they say that “an exact asymptotic analysis of the growth behaviour of the sequence remains a challenge”, give the lower bound of their Theorem 12, and quote the conjectured equivalent $p_n \sim (0.695688 \dots)(0.898118 \dots)^n n!$. Their conclusion (p. 37) lists among open challenges “the existence of closed-form formulas” and says that “the optimal cost of the computation of pop-stacked permutations is still open”. Part I settles one instance of the first (a closed form for the exponential generating function of the total count) and improves the operation count for the second without settling optimality (Section 33, item 5).

Patel’s proof of Conjecture 2, and priority. Part III, Section 30, credits a public proof of Conjecture 2 by Shivam Patel on MathDB [11], posted 20 August 2026, and calls its own proof an alternate one; it makes no first-resolution claim. At intake (6 October 2026, earlier the same day) the page returned HTTP 403. At the write it was reachable, and the write read it in full. The page shows the conjecture with the credit to Claesson, Guðmundsson and Pantone, a site-generated “Progress summary” marked “Claimed solved” that says the posted proof “has not been independently verified”, and one solution by Shivam Patel, whose page metadata give posting and editing times 2026-08-20 14:02:52 and 14:04:08 UTC, as Part III records. The argument encodes a pop-stacked permutation with k runs as a surjective word on $[k]$ in which the occurrence intervals of every two adjacent letters overlap; it applies inclusion–exclusion over the two disjoint ways each adjacency can fail, which turns each term into a count of words respecting a poset P on $[k]$ (a forest of oriented path edges); it classifies such words by the order of first occurrences, which gives $x^k \sum_{\sigma} \prod_{j=1}^k (1 - m_j x)^{-1}$ over linear extensions σ , with m_j the number of maximal elements of the first j letters; since $m_j \leq j$, every denominator divides $\prod_i (1 - ix)^{k-i+1}$; and it evaluates $\lim_{x \rightarrow \infty} F_k(x) = -1$ by reading $\sum_{\sigma} \prod_j 1/m_j$ as the total probability of removing a uniformly random maximal element step by step, so that the numerator has exact degree $k(k+1)/2$ and the stated leading coefficient. The write checked each step and found the argument correct and complete for the denominator, the degree and the leading coefficient, in agreement with Part III’s assessment. Patel also proves $N_k(1) = 2(-1)^{k(k+1)/2-1} \prod_{j=1}^{k-1} j!$ for $k \geq 2$ (only the two consistently oriented full chains have a pole of order k at $x = 1$); the write checked this against the published $N_1, \dots, N_5$ (1, 2, −4, −24, 576). Priority, as far as the write can see: Patel’s proof (20 August 2026) precedes Part III (3 October 2026); MathDB’s status for the problem is “Claimed solved” (“CLAIMED SOLVED

- unverified” in the page data), with the note that the posted proof “has not been independently verified”; no peer-reviewed proof is known to the write. Part III’s proof is different (a degree filtration of the bivariate formula) and stands on its own.

The OEIS entries

The write read both entries on 6 October 2026. **A307030** [9] (revision #63 of November 18, 2021, the revision Parts I and II cite), author Cyril Banderier (March 20, 2019): “Number of permutations of $[n]$ with overlapping adjacent runs”, offset 1 (so the entry starts at p_1 ; the Parts add $p_0 = 1$), with a b-file of $n \leq 450$ by Bjarki Ágúst Guðmundsson, links to the papers above and to Defant–Williams, and no formula or asymptotic field. Part I’s recurrence (8.1), run by the write, reproduces all 450 b-file terms, with divisibility by six at every step. **A309993** [10] (revision #21 of August 29, 2019), author Bjarki Ágúst Guðmundsson (August 26, 2019): the triangle $T(n, k) = p_{n,k}$ for $1 \leq k \leq n$, row sums A307030, a b-file of 5050 terms ($n \leq 100$), and the column generating functions for $k \leq 5$, which are the formulas on p. 10 of [2] that Part III uses as fixtures. Neither entry states an asymptotic formula. The b-file of A309993 was not checked by Part III or by the write. Nothing was submitted to the OEIS.

The inverses and the transseries volume

The repository’s transseries volume [13] has a Part 0 apparatus for inverting rapidly growing functions. Its symbols are subscripted “vol” here where they meet this report’s letters ($\kappa_{\text{vol}}, d_{\text{vol}}, v_{\text{vol}}, X_{\text{vol}}, L_{\text{vol}}, h_{\text{vol}}, f_{\text{vol}}, R_{\text{vol}}$). Statement by statement (Part III has no inverse):

- Parts I and II, the initializer $x_0 = L/W(L/(e\rho))$ (Part II, (19.7); Part I, Section 11, with L_y for $L = \log y$), the root of $x_0 \log(x_0/(e\rho)) = L$: an *instance* of the factorial core **p0:prop:factorial-core** with $\kappa_{\text{vol}} = 1$, $d_{\text{vol}} = -(1 + \log \rho)$ and target $L_{\text{vol}} = L > 0$. That proposition gives $v_{\text{vol}} = W_0(Le^{-1-\log \rho}) = W_0(L/(e\rho))$, which is Part I’s u , and $X_{\text{vol}} = e^{v_{\text{vol}}+1+\log \rho} = e\rho e^{v_{\text{vol}}}$; since $v_{\text{vol}}e^{v_{\text{vol}}} = L/(e\rho)$, $X_{\text{vol}} = L/v_{\text{vol}} = x_0$, and $\kappa_{\text{vol}} \log X_{\text{vol}} + d_{\text{vol}} = v_{\text{vol}} > 0$, the side condition. Its slope formula gives $\Phi'_0(x_0) = v_{\text{vol}} + 1$, which is Part I’s $u + 1$ and Part II’s $u = \log(x_0/\rho)$ (so Part II’s u is Part I’s u plus one, the difference the intake found numerically).
- Part I, the one-step correction (11.4): exactly the first-order term $X - \varrho(X)/\Phi'_0(X)$ of the perturbation-operator series **p0:eq:operator-series** (**p0:prop:operator-form**) about the factorial core $\Phi_0(x) = x \log(x/(e\rho))$, with $X = x_0$, $\Phi'_0(x_0) = u + 1$ and $\varrho(x) = \frac{1}{2} \log x + \log(C\sqrt{2\pi}) + O(1/x)$, the difference $\mathcal{H}(x) - \Phi_0(x)$ by Stirling’s formula. The error bound $O(1/(x_0u))$ is Part I’s own argument; the proposition is a formal series. An *analogue*, not an instance of a theorem of the volume. The same holds for Part II’s (19.8), which is the same formula.
- Part II, the recursion (19.10) for the coefficients d_m : an *instance* of **p0:thm:core-reversion**, with the volume’s quadratic part $h_{\text{vol}} = 0$. Take for R_{vol} the field $\mathbb{Q}(u, b)$ of rational functions in two indeterminates (a commutative $\mathbb{Q}$ -algebra), $\Lambda = u$, the volume’s t equal to Part II’s t , and the unknown $E = d - d_0$ (the volume’s second variable). Since $ud_0 + b = 0$, Part II’s (19.9) reads $\mathcal{F}(d_0 + E, t; u, b) = uE + f_{\text{vol}}(t, E)$ with

$$f_{\text{vol}}(t, E) = \sum_{k \geq 2} \frac{(-1)^k (d_0 + E)^k t^{k-1}}{k(k-1)} + \frac{1}{2} \log(1 + t(d_0 + E)) + \sum_{j \geq 1} \frac{B_{2j} t^{2j-1}}{2j(2j-1)(1 + t(d_0 + E))^{2j-1}}.$$

Every term carries a factor t , and each coefficient of t^m involves finitely many terms, so $f_{\text{vol}} \in tR_{\text{vol}}[[t, E]]$. Hence `p0:eq:core-master` holds with these data, and its triangular recurrence `p0:eq:core-triangular`, $e_m = -u^{-1}[t^m]f_{\text{vol}}(t, E_{<m})$, is Part II's (19.10), because $[t^m]$ of $uE_{<m}$ vanishes; so $e_m = d_m$ for $m \geq 1$. The theorem gives the formal series; the asymptotic statement (19.12), with its error $O(x_0^{-m-1}/\log x_0)$ and the location of the root, is Part II's own. (Part II's expansion of $H(x_0 + d) - L$ into (19.9) was checked by the write: $(x_0 + d)\log(1 + td) = d + \sum_{k \geq 2} (-1)^k d^k t^{k-1}/(k(k-1))$ with $t = 1/x_0$.)

- Part I's Proposition 11.1 and Part II's Theorem 19.2 (the same statement, proved the same way): the two-ceiling bracket is proved directly at the integers from the log-Gamma model and the error bound $M\theta^n$, without an interpolation of the sequence; *not* an instance of `p0:thm:staircase`. When the two ceilings agree they determine $N(y)$, the same conclusion as item (2) of that theorem; both Parts' warnings that an asymptotic root cannot simply be rounded, and Part II's $x - x_0 \rightarrow -\frac{1}{2}$ against a bare ceiling of x_0 , describe the failure mode of its items (2) and (5).

Neither Part claims novelty for the inversion mechanics. The pole sums of Part I's (6.3) are finite and involve no inversion.

Notation across the three Parts

Common to all three Parts with the same meaning: p_n (A307030, $p_0 = 1$), $P(z)$ for the univariate EGF (Part III's $P(z, 1)$), ρ , the states $I = (0, 1)$ and $X = I \sqcup \Delta$, lo, hi, w_z , A_z , λ , spr, B_{2j} (Bernoulli numbers), Γ , ψ , W (the positive real Lambert branch). Part II's amplitude C is Part I's C , and Part II's ρ is Part I's $\rho = \rho_0$ (proved in a note in Section 14). The symbols below change meaning between the Parts, or inside one; each Part fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

Symbol	Meanings
$K, J, \mathcal{K}$	Part I and III: $K(s, t)$ the overlap kernel, $\mathcal{K}$ its operator. Part II: the same kernel is J . Parts I and III: J_0 an entire function, J_1 a cumulative integral. Part III: K_n the run count of a uniform pop-stacked permutation. <i>Tempting false reading</i> : Part III's K_n is a random variable, not the kernel; Part II's J is not Part I's J_0 .
Δ, D	Parts I and III: Δ the interval states; Part II calls them D . Part I: $D(z)$ the denominator (1.4), $D_{\pm}$ half-angle factors. Part II: $D(z)$ the finite matrix of the Fredholm reduction (Section 17), $D_r = M_{w'_r/w_r}$. Part III: $D(z, u) = BH - QJ_0$, $D_{\pm}$, and $D_k(x)$ the fixed-run denominator. <i>False reading</i> : Part II's D in (14.5) is a set, not a denominator.
μ, ν	Parts I and II: μ the measure on X . Part III: ν is that measure and μ the mean slope (21.6); μ_L, μ_U its bounds. Claesson–Guðmundsson–Pantone: μ the dominant pole ρ (front matter only).
$H, \mathcal{H}$	Part I: $H(z)$ entire, locally $\cos(zr/4)$ (1.2); $\mathcal{H}(x)$ the log-Gamma model (11.1). Part II: $H(x)$ that model (19.1); $\mathcal{H}$ the Hilbert space. Part III: $H(z, u)$ entire; $H_c(t)$ the cumulant exponent (27.1). <i>False reading</i> : Part II's H is Part I's $\mathcal{H}$, not Part I's H .
N	Part I: $N(z)$ the numerator (1.4); $N(y)$ the threshold. Part II: $N(y)$ the threshold. Part III: $N(z, u)$ the phase numerator, N_0 , and $N_k(x)$ the fixed-run numerator.
u	Part I: $u(x) = U'(x)$ in the boundary problem; a jet variable in Section 7; $u = W(L_y/(e\rho))$ in Section 11. Part II: $u = \log(x_0/\rho)$, which is Part I's u plus one. Part III: the run marker. <i>False reading</i> : Parts I and II's u in the inverse differ by exactly one.
v	Part I: $v = V'$; a jet variable. Part II: $v = v_{\rho}$ the Perron eigenfunction. Part III: v the variance slope (21.7); v_L, v_U .

Symbol	Meanings
q, θ, η	Part I: q_n in (8.1); $\theta = \rho/R$. Part II: $q = \rho/R$ (Part I's θ). Part III: $q = V'$ in (23.4), $q = e^z$ as a formal variable in Section 30; $\theta = zr/4$, and θ a frequency in Section 29; η the error ratio.
r, R	Parts I and III: $r = \sqrt{4e^z - 1}$, resp. $\sqrt{Q}$. Part II: $r > 0$ a real parameter. All: $R > \rho$ a radius. Part III: $\mathcal{R}$ a ring, R_1, R_2 residuals, R_δ .
T	Part I: $T = P - 1$, T_n binomial sums, T_{200} a Taylor polynomial. Part II: $T_r = M_{\sqrt{w_r}}$. Part III: $T_0 = P - 1$, $T = e^\rho$, T_1, T_{200} .
L	Part I: L the lower rational endpoint (10.1); $L = rc + (2E + 1)s$ in Section 4; $L_y = \log y$. Part II: $L = \log y$. Part III: L, U the endpoints; L_0, L_1 .
C	Parts I and II: the amplitude (the same number); Part I: C_ζ, C_L, C_U . Part III: $C(u)$ the moving amplitude; C a generic constant in (29.2).
B, b	Part I: $B(z) = 2E + 1$; B_1, B_2 resultant coefficients; $B_z = \mathcal{K}M_{ w_z }$. Part II: $B_z = I - E_z$; $b = \frac{1}{2} \log(2\pi x_0) + \log C$. Part III: $B(z, u)$; $b = 2ue^{z/2}$; b_z and B_z^{abs} .
A, E	All: A_z the operator. Part I: A a coefficient in (4.1), $A = (z + 2)p^2$ in Section 7; $E = e^{z/2}$, E_n sums. Part II: $E_z = A_z - F$; E a set. Part III: A in (24.1) and (30.2); $E = e^{z/2}$; $\mathbb{E}$ expectation.
$F, \mathcal{F}, G$	Part I: F a set; $\mathcal{F}$ the resultant (7.1); G_n sums. Part II: F a finite-rank operator; $\mathcal{F}$ the formal inverse series (19.9); $G(z)$ a remainder. Part III: F a set; $F_k(x)$ the fixed-run OGF; $G(z, u)$, $G_c(t)$. <i>False reading</i> : Part II's $\mathcal{F}$ is not Part I's resultant.
h	Part I: h the resolvent function, h_* . Part II: h the right eigenvector. Part III: h , $h_0 = 1 + u(e^z - 1)$, h_z , and $h(\theta)$. The transseries volume's h is written h_{vol} .
d	Part II: d_ρ (14.5) and d_m (19.10). Part III: $d = \sin \theta$. The volume's d is written d_{vol} .
β, γ, δ	Parts I and II: β the Perron eigenvalue of the comparison operator on $ z = \rho$ (Part III calls it b_z). Part III: β, γ the constant terms (27.4); δ a frequency cutoff, and $\delta = \sqrt{(1 - 2u)^2 - 4u^2q}$ in Section 30. Parts I and II: $\delta(x)$ the bracket width. Part I: $\delta_{n,0}$.
s, S	Part I: s_k Stirling coefficients; $s = \sin(w/2)$; S_t ; $S = 10^{120}$; S_n sums. Part II: S_r ; s_k . Part III: $s = 2u - 1$; S_t ; $S = 10^{120}$.
ϕ, Φ, α	Part I: $\phi(z) = zr/4 + \alpha(z)$ (5.1). Part III: the same ϕ at $u = 1$, and $\Phi(z, u)$. The volume's core is written Φ_0 (front matter only).
M, Q, κ	Parts I and II: M a constant in the log error; M_f multiplication. Part II: $M = \max \text{hi}$ in Section 16. Part III: $M = xa_0 - 1$. Part I: $Q = 4e^z - 1$. Part II: $Q_z = JM_{ w_z }$. Part III: $Q(z, u)$; $Q_0 = 4T - 1$; $\kappa = u + u^2(e^z - 1)$ (the volume's κ is κ_{vol}).
x, X, ℓ	All: X the state space; Part I also uses X for e^z in Section 7. Part II: x the real root of $H(x) = L$, x_0 . Part III: $x = \rho$ in Sections 27–28, and x the OGF variable. Part II: ℓ the left eigenvector (Part I's ℓ_*); ℓ a run length elsewhere.

Provenance and merge decisions

Three manuscripts of the session bundle of Reports 1–243, all dated 3 October 2026. Their author lines read “Report 164”, “Report 161” and “Report 166”; none names an author, a tool or an addressee, none pins a repository commit, none carries “prepared for private review” wording, and their PDF author fields are empty. They arrived in commit 60f54ea06 and were placed by 602e5bd0f (batch 108, cluster 108-POP).

- **Part I**, bundle Report 164, archive `Pop_Stacked_Permutations_Exact_Generating_Function_and_Poles_Source.zip` (464,784 bytes; 16 files; `Report164.tex`, 856 lines, 15 pages); the base. Contributes the explicit formula, the pole theory, non-D-finiteness, differential algebraicity, the recurrence, the closed amplitude and the certificate.
- **Part II**, bundle Report 161, archive `Pop_Stacked_Permutations_Asymptotics_and_Inver`

`ses_Source.zip` (517,332 bytes; 15 files; `Report161.tex`, 806 lines, 13 pages). Contributes the operator proof with the Fredholm continuation, the elementary bracket for ρ , the all-orders inverse, the endpoint recurrence as used by its companion, and the count data that Part I's fixture is derived from.

- **Part III**, bundle Report 166, archive `Pop_Stacked_Permutations_Run_Distribution_Source.zip` (579,822 bytes; 19 files; `Report166.tex`, 901 lines, 16 pages). Contributes the bivariate formula, the run statistics, and the alternate proof of Conjecture 2.

One chain between the packages. Part I's count fixture `data/164-egf-data-counts_p0_p70.json` names "Report161/data/counts.json" as its source and records a SHA-256 beginning `d42dae1b` (recorded also in its `verification_n70.json` and in its companion); that is the SHA-256 of Report 161's delivered `data/counts.json`, shipped here as `data/161-operator-data-counts.json` (checked at the write). So $p_{26}, \dots, p_{70}$ in Part I's fixture come from Report 161's endpoint-recurrence computation, and only $p_1, \dots, p_{25}$ are posted OEIS terms, as the fixture itself says.

Where the merge had to choose:

1. *Base and order.* Report 164 first, as the base (see the Guide); 161 second; 166 third. The alternative of basing on 166's bivariate formula was not taken, because 166 has none of 164's pole results.
2. *Repeated arguments.* The three Parts prove the same run-state normalization and resolvent, the same Perron lemma and the same strict comparison on $|z| = \rho$, and Parts I and III use the same rational certificate for ρ . The texts are independent paraphrases with different detail (Part II writes out the Fredholm continuation; Part III carries the marker u). Each Part's proof is complete only with its own version, so each is printed as delivered, with `[write]` pointers to the other places; nothing was dropped.
3. *Front matter, numbering and bibliography.* The three title blocks and abstracts are printed at the heads of their Parts, with each manuscript's opening scope paragraph; one table of contents replaces three. The three bibliographies are merged into one, keeping every detail of every entry (where the Parts give different links or locators, all are kept and attributed); the two entries keyed `oeis` became `oeisA307030` (Parts I and II) and `oeisA309993` (Part III). The write added entries for the transseries volume.
4. *Write additions.* This front matter, the reading-conventions tables of the Parts, Remark 5.1, Section 33, the dated notes, labels for Part I's Lemma 3.1 and Proposition 11.1, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 108, 6 October 2026) all three manuscripts were read in full and no mathematical step was found false. Checked there: the trigonometric form (1.6) against $p_0, \dots, p_{25}$ (60 digits); ρ , C and $p_{70}/(70! \rho^{-70}) = C$ to 18 digits; $\rho_0, \dots, \rho_{15}$ from the phase equation; the three printed complex pairs as zeros of D ; Part II's operator argument line by line (left eigenvector $\sqrt{w}v$, the strict-domination identity, λ' , the scalar numerator); μ and v against $-\rho'(1)/\rho$ and the second-order formula by implicit differentiation of $D(z, u) = 0$; Part III's degrees and leading coefficients against the published $N_1, \dots, N_5$; and the companions on copies (all outputs reproduced).

The write (6 October 2026) re-read Claesson–Guðmundsson–Pantone (arXiv version 1 and the author PDF, whose SHA-256 is the one recorded by Parts I and III) and Asinowski–Banderier–Hackl (Sections 2.2–2.4 and the conclusion), the OEIS entries, the authors’ repository file of estimates, and Patel’s MathDB page; proved the interval statements of Remark 5.1; recomputed $\rho_0, \dots, \rho_{19}$, the eight repository complex pairs and C in 220-digit arithmetic; ran Part I’s recurrence against the 450 b-file terms of A307030; recomputed μ, v, β, γ by numerical differentiation of Part III’s $\rho(u)$ and $C(u)$, and the exact mean and variance of K_n for $n \leq 40$ from the delivered rows; checked the identities of the preceding sections; and reran the companions on fresh extractions of the three archives (Report 161’s `counts`, `verify` and `sources`, Report 164’s `verify --n 70` and `certify`, Report 166’s full `run_checks.py` check): every output reproduced the shipped file byte for byte after removing carriage returns.

What was *not* done: the analytic proofs (the sufficiency lemmas, the half-angle cancellations, the Perron and Fredholm arguments, the complex-uniform transfer, the local limit estimate, the filtration argument) were read and their displayed algebra spot-checked, but they are the manuscripts’ proofs, not an independent verification. The positions of the nonreal poles are numerical. Defant–Williams [5], Choi–Sun [4], Ehrenborg–Kitaev–Perry [6], Kato [8] and Flajolet–Sedgewick [7] were not re-read by the write, nor the journal version of [3]. The PDF and ZIP release tools were not run (they need POSIX facilities).

Relation to neighbouring reports and formal projects

No other placed report of the repository treats A307030, A309993, pop-stack sorting, or the papers of Claesson–Guðmundsson–Pantone and Asinowski et al. (repository search, 6 October 2026), and none uses the integral-operator method of Ehrenborg, Kitaev and Perry that all three Parts credit. The nearest neighbour by subject is **a113226–vincular-avoiders** (exponential generating functions and asymptotics of a permutation class defined by adjacency conditions); it shares no lemma with this report. The transseries volume [13] is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats pop-stacked permutations, and no statement of this report is formalized.

Part I

Exact generating function for pop stacked permutations: poles, differential algebraicity and faster enumeration

Bundle Report 164, archive `Pop_Stacked_Permutations_Exact_Generating_Function_and_Poles_Source.zip`, delivered as `Report164.tex` (15 pages), the base of the merge. Delivered title block “Exact generating function for pop stacked permutations / Poles differential algebraicity and faster enumeration / Report 164 / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. Remark 5.1 and the labels of Lemma 3.1 and Proposition 11.1 are added in the write. The opening paragraph “Scope and reproduction” is the manuscript’s own.

Abstract of this Part (as delivered). Let p_n count permutations in the image of one pass through a pop stack, with $p_0 = 1$. We derive an explicit exponential generating function $P(z) = \sum_{n \geq 0} p_n z^n / n!$ for OEIS A307030. Its branch-free expression is a quotient of entire functions. A cumulative integral boundary problem and an explicit half-angle cancellation establish the formula from the run-overlap characterization, rather than from fitted coefficients.

Every genuine complex pole is simple and has an explicit residue. There are infinitely many positive real poles, proving that P is not D-finite and (p_n) is not P-recursive. Nevertheless P is differentially algebraic, with an order-at-most-two polynomial annihilator given by a small resultant. An exact Riccati equation computes $p_0, \dots, p_N$ in $O(N^2)$ arithmetic operations and $O(N)$ scalar storage, improving the $O(N^4)$ arithmetic-operation total-count algorithms in the cited papers.

A self-contained positive-operator argument isolates the unique dominant pole ρ . We obtain its exact amplitude C , rationally certify narrow intervals for both, and prove finite-radius pole expansions. Fixed-order Stirling and rounding-safe inverse consequences follow. The numerical pole estimates and leading constants are prior art; the contribution here is the explicit formula and proofs. No effective numerical complex gap, infinite pole-sum convergence, bit-complexity, optimality, or unrestricted priority claim is made.

Scope and reproduction. This is a standalone derivation. The overlap characterization and earlier counting algorithms are attributed in Section 12. A bounded standard-library companion independently expands the entire quotient, checks the integer recurrence, and certifies the constants with rational inequalities. Its checks do not replace the proofs. All computed public commands write deterministic JSON to standard output only. The source package includes the report, reproducible PDF builder, fixtures, tests, and checked output certificates. The existing Report 161 is unchanged.

Reading conventions of Part I `[write]`

p_n, P, N, D	A307030 with $p_0 = 1$; its EGF; numerator and denominator (1.4)
Q, E, B, H, J_0	$4e^z - 1$, $e^{z/2}$, $2E + 1$, and the entire series (1.2)–(1.3)
$K, w_z, A_z, \mu, X = I \sqcup \Delta$	overlap kernel (Part II’s J), weights, operator, measure, states (Δ is Part II’s D)
U, V, u, v, I_0, J_1	endpoint cumulatives and the boundary problem of Section 3
$r, \phi, \alpha, \rho_k, \rho$	$\sqrt{4e^z - 1}$, the phase (5.1), the positive poles, $\rho = \rho_0$
C, C_ζ, θ, R	amplitude (equal to Part II’s C), pole coefficients, $\theta = \rho/R$ (Part II’s q)
$\mathcal{H}, N(y), u, x_0, L_y$	log-Gamma model (Part II’s H), threshold, $u = W(L_y/(e\rho))$ (Part II’s u minus one), initializer, log y
L, U, C_L, C_U	rational endpoints of the certificate (also Part III’s L, U)

1 The explicit generating function

A pop-stack pass reverses each maximal decreasing run and concatenates the resulting increasing blocks. A permutation in its image is *pop-stacked*. The first values, including the empty permutation, are

$$1, 1, 1, 3, 11, 49, 263, 1653, 11877, 95991, 862047, 8516221.$$

The established characterization [1, 2] says that consecutive maximal ascending runs, with endpoints (a, b) and (c, d) , satisfy

$$a < d, \quad c < b. \quad (1.1)$$

A singleton has equal endpoints. Thus a singleton may sit strictly inside the range of an adjacent run, but two singleton runs cannot be adjacent. The second inequality enforces the descent between maximal runs; the first is the extra image condition.

Put

$$Q(z) = 4e^z - 1, \quad E(z) = e^{z/2}, \quad B(z) = 2E(z) + 1,$$

and define the entire functions

$$H(z) = \sum_{k \geq 0} \frac{(-1)^k z^{2k} Q(z)^k}{16^k (2k)!}, \quad (1.2)$$

$$J_0(z) = \frac{z}{4} \sum_{k \geq 0} \frac{(-1)^k z^{2k} Q(z)^k}{16^k (2k+1)!}. \quad (1.3)$$

Factorial denominators give locally uniform convergence. Set

$$N(z) = e^{-z} Q(z) (H(z) + B(z) J_0(z)), \quad D(z) = B(z) H(z) - Q(z) J_0(z). \quad (1.4)$$

Theorem 1.1. *The exponential generating function of pop-stacked permutations is*

$$\boxed{P(z) = \frac{N(z)}{D(z)}}. \quad (1.5)$$

This denotes a meromorphic function on $\mathbb{C}$, with removable common zeros filled in. Near zero, write $r = \sqrt{4e^z - 1}$ with $r(0) = \sqrt{3}$; then

$$P(z) = e^{-z} r \frac{(2E + 1) \sin(zr/4) + r \cos(zr/4)}{(2E + 1) \cos(zr/4) - r \sin(zr/4)}. \quad (1.6)$$

The global expression (1.5) uses no square-root branch. The genuine pole set is exactly

$$\{\zeta \in \mathbb{C} : D(\zeta) = 0, \ Q(\zeta) \neq 0\}. \quad (1.7)$$

Every member is simple, and

$$\text{Res}(P, \zeta) = -\frac{2e^{-2\zeta}(4e^\zeta - 1)}{\zeta + 2}. \quad (1.8)$$

There are infinitely many positive poles. The EGF is not D-finite, and the counts are not P-recursive, although the EGF is differentially algebraic over $\mathbb{Q}(z)$.

The proof occupies Sections 2–7. The exact recurrence and arithmetic complexity appear in Section 8; the leading asymptotic and constants appear in Sections 9–10.

[write] **6 October 2026.** Theorem 1.1 proves the predictions of Claesson, Guðmundsson and Pantone [3, Section 3.3] that the poles are simple, that there are infinitely many singularities, and that neither the exponential nor the ordinary generating function is D-finite (front matter, “The predictions of Claesson, Guðmundsson and Pantone”). Part III’s Theorem 21.1 reduces to (1.5) at $u = 1$, where its $Q = 4e^z - 1$, $B = 2e^{z/2} + 1$ and $h_0 = e^z$; its derivation through the Robin problem (23.5) is therefore a second derivation of this formula. Part III proves none of the pole statements.

2 Mixed run states and exact normalization

Let

$$I = (0, 1), \quad \Delta = \{(a, b) : 0 < a < b < 1\}, \quad X = I \sqcup \Delta, \quad \mu = dx \sqcup da db.$$

A state $x \in I$ is a singleton with $\text{lo}(x) = \text{hi}(x) = x$; a state $(a, b) \in \Delta$ has $\text{lo} = a, \text{hi} = b$. Define

$$K(s, t) = \mathbf{1}\{\text{lo}(s) < \text{hi}(t), \text{lo}(t) < \text{hi}(s)\}, \quad (2.1)$$

$$w_z(x) = z, \quad w_z(a, b) = z^2 e^{z(b-a)}, \quad (2.2)$$

$$(A_z f)(s) = \int_X K(s, t) w_z(t) f(t) d\mu(t). \quad (2.3)$$

Work, for example, on $L^2(X, \mu)$. The kernel is bounded, $\mu(X) = 3/2$, and the weights are uniformly bounded on compact z -sets. Thus A_z is compact, entire in operator norm, and $A_z = O(|z|)$ near zero.

Here is the factorial normalization. Represent n distinct labels by independent coordinates in $(0, 1)$. Each of the $n!$ rank orders has volume $1/n!$. A run of length $\ell \geq 2$ with extreme coordinates $a < b$ has ordered interior-coordinate volume $(b - a)^{\ell-2}/(\ell - 2)!$. Its summed weight is

$$\sum_{\ell \geq 2} z^\ell \frac{(b - a)^{\ell-2}}{(\ell - 2)!} = z^2 e^{z(b-a)}.$$

A singleton contributes $z dx$. The product of kernels enforces exactly the adjacent-run conditions (1.1); there is neither an extra run-order factor nor an extra factorial. For instance,

$$\int_X w_z d\mu = z + z^2 \int_0^1 (1 - t) e^{zt} dt = e^z - 1,$$

the EGF of the unique increasing permutation of each positive size.

Summing over the number of runs, with absolute convergence for small $|z|$, now proves

$$h = (I - A_z)^{-1} \mathbf{1}, \quad P(z) = 1 + \int_X w_z h d\mu. \quad (2.4)$$

The pairing is bilinear; no complex conjugate is inserted.

[write] The same normalization and resolvent are proved in Part II, Section 15 (with J for K and D for Δ), and in Part III, Section 22 (every run weight multiplied by u , measure ν); (2.4) is (15.2) and the case $u = 1$ of (22.4). The overlap characterization (1.1) is proved in Part II, Lemma 15.1.

3 Cumulative boundary problem

For small z , define the endpoint cumulatives

$$U(x) = \int_{\text{hi}(s) \leq x} w_z(s) h(s) d\mu(s), \quad V(x) = \int_{\text{lo}(s) < x} w_z(s) h(s) d\mu(s),$$

$$T = \int_X w_z h d\mu = P - 1.$$

Endpoint equality sets have measure zero. Hence

$$U(0) = V(0) = 0, \quad U(1) = V(1) = T. \quad (3.1)$$

Complementing the two nonoverlap regions in (2.1) gives

$$h(x) = 1 + V(x) - U(x), \quad h(a, b) = 1 + V(b) - U(a). \quad (3.2)$$

These identities first yield continuous representatives and then the smoothness used below. Introduce

$$I_0(x) = \int_0^x e^{z(x-a)} U(a) da, \quad J_1(x) = \int_x^1 e^{z(b-x)} V(b) db.$$

Direct differentiation of the cumulatives gives

$$U' = ze^{zx}(1 + V) - zU - z^2 I_0, \quad I_0' = U + zI_0, \quad (3.3)$$

$$V' = zV + ze^{z(1-x)}(1 - U) + z^2 J_1, \quad J_1' = -V - zJ_1. \quad (3.4)$$

For example, differentiating the upper endpoint of the interval part of U gives $z^2 \int_0^x e^{z(x-a)} (1 + V(x) - U(a)) da$; combining this with the singleton term $z(1 + V(x) - U(x))$ gives (3.3). Differentiating again and substituting eliminates the integral terms:

$$U'' = ze^{zx} V', \quad V'' = -ze^{z(1-x)} U'. \quad (3.5)$$

Writing $u = U'$ and $v = V'$ yields

$$u' = ze^{zx} v, \quad v' = -ze^{z(1-x)} u, \quad u(0) = z, \quad v(1) = z. \quad (3.6)$$

The final boundary condition uses $U(1) = V(1)$ in (3.4). For $z \neq 0$, elimination gives the constant-coefficient problem

$$u'' - zu' + z^2 e^z u = 0, \quad u(0) = z, \quad u'(1) = z^2 e^z. \quad (3.7)$$

At zero the homogeneous scalar problem is $u'' = 0, u(0) = u'(1) = 0$; its only solution is zero. Analytic fundamental matrices therefore give local uniqueness for (3.7). The first-order problem (3.6) is likewise locally unique, including $z = 0$.

Lemma 3.1 (Sufficiency of the boundary problem). *The locally unique solution of (3.6), integrated from zero, reconstructs precisely the resolvent in (2.4).*

Proof. The reflected pair $(v(1-x), u(1-x))$ solves the same equations and boundary data. Uniqueness gives $u(x) = v(1-x)$. Therefore the reconstructed $U(x) = \int_0^x u$ and $V(x) = \int_0^x v$ satisfy $U(1) = V(1)$. Define I_0, J_1 from these functions and put

$$F_1 = u - ze^{zx}(1 + V) + zU + z^2 I_0,$$

$$F_2 = v - zV - ze^{z(1-x)}(1 - U) - z^2 J_1.$$

Calculation using (3.6) gives $F'_1 = zF_1$ and $F'_2 = -zF_2$. Their boundary values are $F_1(0) = 0$ and $F_2(1) = z(U(1) - V(1)) = 0$. Both vanish. Consequently the endpoint cumulatives of the h defined by (3.2) have derivatives U', V' and initial values zero; they are exactly U, V . Complementing nonoverlap regions gives $h = 1 + A_z h$. The Neumann inverse is unique near zero, completing the sufficiency proof. $\square$

This step retains the endpoint condition that could otherwise be lost by differentiating. No global uniqueness of the unreduced boundary problem is asserted at exceptional complex parameters.

4 Explicit solution and the canceled factor

Choose $r = \sqrt{4e^z - 1}$ locally at zero and put $w = zr/2$. Solving (3.7) gives

$$u(x) = e^{zx/2}(z \cos(wx) + A \sin(wx)), \quad (4.1)$$

$$\frac{A}{z} = \frac{2E - \cos w + r \sin w}{r \cos w + \sin w}. \quad (4.2)$$

Integrate the differential equation and use $P = 1 + \int_0^1 u$ to obtain

$$P = \frac{zu(1) - z^2/2 + Aw}{z^2 E^2}. \quad (4.3)$$

Substitution, with $\sin^2 w + \cos^2 w = 1$, simplifies this to

$$P = e^{-z} r \frac{2E - \cos w + r \sin w}{r \cos w + \sin w}. \quad (4.4)$$

All identities with apparent division by z are first obtained for small nonzero z and then continued to zero.

To display the cancellation, set $c = \cos(w/2)$, $s = \sin(w/2)$ and

$$D_+ = (2E + 1)c - rs, \quad D_- = (2E - 1)c + rs, \quad L = rc + (2E + 1)s.$$

The identity $r^2 = (2E - 1)(2E + 1)$ gives, by multiplication,

$$r(r \cos w + \sin w) = D_+ D_-, \quad (4.5)$$

$$r(2E - \cos w + r \sin w) = D_- L. \quad (4.6)$$

As $D_-(0) = 1$, the common factor cancels near zero. Substituting into (4.4) proves (1.6). In particular, zeros of the full boundary determinant coming solely from D_- are not poles of the scalar counting EGF.

Locally $H = \cos(zr/4)$ and $J_0 = \sin(zr/4)/r$, so (1.6) is exactly (1.5). Both N, D are entire and $N(0) = D(0) = 3$. This proves the branch-free meromorphic continuation and $P(0) = 1$. It does not require a square root or inverse trigonometric function to be defined globally.

5 Positive poles and non D finiteness

For real $z \geq 0$ put $r = \sqrt{4e^z - 1} > 0$ and

$$\alpha(z) = \frac{1}{2} \arccos\left(\frac{1}{2e^{z/2}}\right), \quad \phi(z) = \frac{zr}{4} + \alpha(z). \quad (5.1)$$

The real inverse cosine takes values in $[0, \pi]$. Since $\tan \alpha = r/(2E + 1)$, equation (1.6) becomes

$$P(z) = e^{-z} r \tan \phi(z). \quad (5.2)$$

Differentiation simplifies to

$$\phi'(z) = \frac{e^z(z+2)}{2r} > 0. \quad (5.3)$$

Moreover $\phi(0) = \pi/6$ and $\phi(z) \rightarrow \infty$. Thus for each integer $k \geq 0$ there is a unique positive ρ_k satisfying

$$\phi(\rho_k) = \frac{\pi}{2} + k\pi, \quad (5.4)$$

or equivalently

$$\frac{\rho_k \sqrt{4e^{\rho_k} - 1}}{2} + \arccos\left(\frac{1}{2e^{\rho_k/2}}\right) = (2k+1)\pi.$$

The derivative and prefactor in (5.2) are nonzero there, so all these points are genuine simple poles. This describes positive poles only, without ordering them among complex poles by modulus.

Remark 5.1 ([write]) The printed list of predicted positive poles omits ρ_{12} . Claesson, Guðmundsson and Pantone [3, Section 3.3, p. 10] print fifteen predicted positive singularities of P as truncated decimals: nine with 75 decimals, then one each with 56, 47, 36, 18, 11 and 6 decimals, ending

$$\dots, 5.08176797057\dots, 5.21558801277\dots, 5.453200964209036692, 5.55979961612, 5.659669.$$

Claim. These are, in order, $\rho_0, \dots, \rho_{11}, \rho_{13}, \rho_{14}, \rho_{15}$ of (5.4), and every printed digit is correct; the list omits the positive pole $\rho_{12} = 5.33887671271979\dots$, which lies strictly between its twelfth and thirteenth entries.

Proof. For real $z > 0$ put $E = e^{z/2} > 1$ and $r = \sqrt{4e^z - 1} > 0$. Since $\tan \alpha = r/(2E + 1)$ with $\alpha = \alpha(z) \in (\pi/6, \pi/4)$ (because $1/(2E) \in (0, \frac{1}{2})$), the denominator of (1.6) is

$$D(z) = (2E + 1) \cos \frac{zr}{4} - r \sin \frac{zr}{4} = \sqrt{(2E + 1)^2 + r^2} \cos \phi(z).$$

As $Q(z) > 0$, Theorem 1.1 makes the positive poles exactly the positive zeros of D , that is the points where $\phi(z) \in \frac{\pi}{2} + \pi\mathbb{Z}$; by (5.3) and $\phi(0) = \pi/6$ the k -th of them is ρ_k , the solution of (5.4). At ρ_k , $zr/4 = \phi - \alpha$ lies in $(k\pi + \pi/4, k\pi + \pi/3)$, so k is the integer part of $zr/(4\pi)$ there, and $zr/(4\pi) - k \in (\frac{1}{4}, \frac{1}{3})$. For each printed value s with d decimals the write evaluated $D(s)$, $D(s + 10^{-d})$ and the range of $zr/(4\pi)$ over $[s, s + 10^{-d}]$ in interval arithmetic (`mpmath.iv`, 120 digits, outward rounding; only `exp`, `sqrt`, `cos`, `sin` and `pi` are needed). In all fifteen cases the two values of D have opposite strict signs, so D has a zero in the open interval $(s, s + 10^{-d})$, and the range of $zr/(4\pi)$ lies in $(k + \frac{1}{4}, k + \frac{1}{2})$ for one integer k , which identifies that zero as ρ_k ; the values of k are $0, 1, \dots, 11, 13, 14, 15$. Hence s is the truncation of ρ_k to d decimals. The same computation with $s = 5.3388767127$, $d = 10$, gives a sign change and $k = 12$. Since ϕ is increasing, $\rho_{11} < \rho_{12} < \rho_{13}$, and ρ_{12} is not in the list. $\square$

The positive poles to twelve decimals (rounded; from a 220-digit solution of (5.4), each with $|D(\rho_k)| < 10^{-190}$); “printed” gives the number of decimals in the list of [3]:

k	ρ_k	printed	k	ρ_k	printed
0	1.113439041737	75	10	5.081767970571	47
1	2.417184228723	75	11	5.215588012778	36
2	3.076673197412	75	12	5.338876712720	omitted
3	3.527590791728	75	13	5.453200964209	18
4	3.872438162423	75	14	5.559799616126	11
5	4.152519207830	75	15	5.659669438240	6
6	4.388766437824	75	16	5.753624587263	–
7	4.593300493040	75	17	5.842338823685	–
8	4.773787732301	75	18	5.926376164810	–
9	4.935393550294	56	19	6.006213584191	–

This corrects a list, not a theorem: the authors present experimental estimates and do not say that the list is complete. Their repository file [12], committed on 20 August 2019 (three days before arXiv version 1) and unchanged since, lists twenty positive values, ρ_0 to ρ_{19} , among them 5.33887671271979787441685886230975425489043134007(1); the repository’s twenty values agree with the solutions of (5.4) within the stated uncertainty of their last digits (for $\rho_0, \dots, \rho_3$, whose repository strings have 252–782 decimals, this needs more than the write’s 220 digits; an independent 900-digit check confirms it). So the omission is one of transcription into the paper. The same printed page lists three nonreal pairs to 73 decimals. Newton’s method on D , started from the printed digits, converges in 220-digit arithmetic to zeros of D with $Q \neq 0$, which are simple poles by Theorem 1.1. The printed digits are correct up to decimal place 62 to 66 (the first wrong digits are at places 67 and 64, 66 and 63, 67 and 63 for the real and imaginary parts of the three pairs); from there each printed string is the repository’s string with two consecutive digits deleted, and the repository’s values agree with the zeros to at least 108 decimal places. These complex computations are floating-point, not interval certificates; the scripts are not shipped, and the method is stated here in full. The intake found the omission of ρ_{12} ; the write proved it, and found the complex misprints.

An analytic D-finite germ satisfies a nonzero linear differential equation with polynomial coefficients. At every point where its leading coefficient is nonzero, the standard local linear-ODE continuation theorem excludes singularities. Its finite-plane singularities can thus occur at only finitely many points. The infinitely many distinct poles above prove that P is not D-finite. If (p_n) were P-recursive, then $(p_n/n!)$ would be P-recursive and its ordinary generating function P would be D-finite, a contradiction. The ordinary series $\sum p_n z^n$, understood formally, is not D-finite either.

[write] **6 October 2026.** This proves what Claesson, Guðmundsson and Pantone [3, Section 3.3, p. 10] predicted (“If true, this would imply the non-D-finiteness of both the ordinary and exponential generating functions”), and what Asinowski, Banderier and Hackl [2, Section 2.4, p. 18] report as their conjecture. Part II’s statements that further poles and non-D-finiteness are not proved (Sections 14 and 20) were true of Report 161 and are superseded here; dated notes there.

6 Riccati equation and every complex pole

On a local branch in (5.2), let $a = e^{-z}r$. Then

$$\frac{a'}{a} = \frac{1 - 2e^z}{Q}, \quad a\phi' = \frac{z + 2}{2}, \quad \frac{\phi'}{a} = \frac{e^{2z}(z + 2)}{2Q}.$$

Differentiating $P = a \tan \phi$ yields the exact identity

$$2QP' = 2(1 - 2e^z)P + (z + 2)Q + (z + 2)e^{2z}P^2, \quad P(0) = 1. \quad (6.1)$$

It holds globally by meromorphic continuation. Since $2Q(0) = 6$, it is also a locally unique analytic initial-value equation at zero.

6.1 Removal of the degenerate parameters

All zeros of Q are $z_k = -\log 4 + 2\pi i k$, and they are simple. At such a point $E = (-1)^k/2$. If k is even, $B = 2$, $D = 2$ and $P(z_k) = 0$. If k is odd, locally

$$B = 1 - \sqrt{1 + Q} = -Q/2 + O(Q^2), \quad H = 1 + O(Q), \quad J_0 = z/4 + O(Q).$$

Consequently

$$D = -Q(z + 2)/4 + O(Q^2), \quad N = 4Q + O(Q^2).$$

As $z_k \neq -2$, the singularity is removable and $P(z_k) = -16/(z_k + 2)$. Thus no zero of Q is a pole.

The other potentially degenerate coefficient for a Riccati pole balance is $z = -2$. Put $s = \sqrt{1 - 4/e^2}$, so $0 < s < 1$. Direct evaluation gives

$$D(-2) = (1 + 2/e) \cosh(s/2) - s \sinh(s/2) > 0,$$

because $1 + 2/e > 1 > s \tanh(s/2)$. Hence -2 is not a pole.

6.2 Simplicity residues and the exact pole set

At a genuine pole ζ , the preceding argument gives $Q(\zeta) \neq 0$ and $\zeta \neq -2$. If its order is m , the most singular orders in (6.1) are $m + 1$ from P' and $2m$ from P^2 ; the other terms are of lower order. Thus $m + 1 = 2m$ and $m = 1$. Writing $P = a_{-1}/(z - \zeta) + O(1)$ and comparing coefficients gives (1.8).

Away from $Q = 0$, a zero of D cannot cancel against N . Indeed the two linear combinations in (1.6) have coefficient determinant

$$(2E + 1)^2 + r^2 = 4E(2E + 1) \neq 0.$$

The last inequality follows because $E \neq 0$ and $2E + 1 = 0$ would imply $Q = 0$. Simultaneous vanishing would force $\sin(zr/4) = \cos(zr/4) = 0$, which is impossible. This proves exactly (1.7).

For a genuine pole define

$$C_\zeta = -\frac{\text{Res}(P, \zeta)}{\zeta} = \frac{2e^{-2\zeta}(4e^\zeta - 1)}{\zeta(\zeta + 2)}. \quad (6.2)$$

There is no pole at zero. For every fixed $R > 0$ with no pole on $|z| = R$, there are finitely many poles inside, and subtraction of their principal parts leaves a function analytic on a neighborhood of the closed disk. Cauchy's bound therefore proves

$$\frac{p_n}{n!} = \sum_{|\zeta| < R} C_\zeta \zeta^{-n} + O_R(R^{-n}). \quad (6.3)$$

This is a finite-radius expansion. It does not assert convergence of an infinite sum over all poles, a numerical ordering of nonreal poles, or an effective bound for the constant implicit in the remainder.

7 Differential algebraicity

Use independent jet variables p, u, v and an auxiliary variable X . Define the following polynomials in $\mathbb{Q}[z, p, u, v]$:

$$\begin{aligned} A &= (z+2)p^2, & B_1 &= -8u - 4p + 4(z+2), & C_1 &= 2u + 2p - (z+2), \\ A_2 &= p^2 + 2(z+2)pu + 2A, & B_2 &= -8v - 4u + 4 + B_1, & C_2 &= 2v + 2u - 1. \end{aligned}$$

The Riccati equation states $AX^2 + B_1X + C_1 = 0$ at $X = e^z$, $p = P$, $u = P'$. Differentiating with $X' = X$ gives $A_2X^2 + B_2X + C_2 = 0$ at $v = P''$. Therefore

$$\mathcal{F}(z, p, u, v) = \det \begin{pmatrix} A & B_1 & C_1 & 0 \\ 0 & A & B_1 & C_1 \\ A_2 & B_2 & C_2 & 0 \\ 0 & A_2 & B_2 & C_2 \end{pmatrix} \quad (7.1)$$

satisfies $\mathcal{F}(z, P, P', P'') = 0$. It is not the zero polynomial: at $(z, p, u, v) = (0, 1, 0, 0)$, the two quadratics are $2X(X+2)$ and $5X^2 + 8X - 1$. Their resultant is $2^2(-1)(3) = -12$. Thus P is differentially algebraic of order at most two over $\mathbb{Q}(z)$, despite being non-D-finite. The polynomial condition can have extraneous solution germs; neither minimal order nor uniqueness from this resultant is asserted. Earlier unsuccessful bounded D-algebraic fitting experiments do not establish differential transcendence and do not contradict this annihilator.

8 Exact quadratic arithmetic enumeration

Let $q_n = 4 - \delta_{n,0}$ for $n \geq 0$, with $q_{-1} = 0$. For $n \geq 0$ put

$$\begin{aligned} E_n &= \sum_{k=0}^n \binom{n}{k} p_k, & S_n &= \sum_{k=0}^n \binom{n}{k} p_k p_{n-k}, \\ T_n &= \sum_{k=0}^n \binom{n}{k} 2^{n-k} S_k, & G_n &= \sum_{k=0}^{n-1} \binom{n}{k} p_{k+1}, \end{aligned}$$

where $T_{-1} = G_0 = 0$. Extracting exponential coefficients in (6.1) gives

$$\boxed{6p_{n+1} = 2p_n - 4E_n + 2q_n + nq_{n-1} + 2T_n + nT_{n-1} - 8G_n}, \quad p_0 = 1. \quad (8.1)$$

For example, the left side of (6.1) contributes $6p_{n+1} + 8G_n$, while the three terms on the right contribute $2p_n - 4E_n$, $2q_n + nq_{n-1}$, and $2T_n + nT_{n-1}$.

All inputs at stage n are already available from $p_0, \dots, p_n$ and previous S_k, T_k . Each finite sum has $O(n)$ terms. Computing through p_N thus takes $O(N^2)$ arithmetic operations. The binomial row can be streamed from $\binom{n}{0} = 1$ by $\binom{n}{k+1} = \binom{n}{k}(n-k)/(k+1)$; powers of two can likewise be updated exactly. Storing the three sequences uses $O(N)$ scalar entries. Divisibility by six is a consequence of the EGF identity, and is also checked explicitly by the companion at every stage.

This improves the total-count bounds of $O(N^4)$ arithmetic operations and $O(N^3)$ scalar storage in [3, Section 2] [2, Theorem 11]. The comparison is in the constant-cost arithmetic model used there. Integers grow with n ; no $O(N^2)$ bit-complexity, wall-clock guarantee, optimality, or corresponding run-refined bound is claimed.

[write] **6 October 2026.** Locations, checked by the write: Claesson–Guðmundsson–Pantone state “ $O(N^4)$ time using $O(N^3)$ memory” in Section 2 (p. 5), assuming $O(1)$ arithmetic. In Asinowski–Banderier–Hackl, Theorem 11 (p. 16) states the run-refined recurrence (their (9)) and, in its last sentence, that p_n “can be computed with $\sim n^4/8$ time-complexity and $\sim n^3/3$ memory-complexity”; its proof and the discussion after it (pp. 16–17) restate this as $O(N^4)$ arithmetic operations with $O(N^3)$ simultaneous allocations and give $O(N^5)$ operations for all $p_{n,k}$. Part I’s citation of Theorem 11 is therefore accurate. (Corrected after the independent check of 6 October 2026. The write first read: “In Asinowski–Banderier–Hackl the total-count bounds ($O(N^4)$ arithmetic operations, $O(N^3)$ simultaneous allocations, with leading terms $N^4/8$ additions and $N^3/3$ allocations) are in the discussion following Theorem 11 (pp. 16–17); Theorem 11 itself is the run-refined recurrence (their (9)), which computes all $p_{n,k}$ in $O(N^5)$ operations.” That placed the total-count costs outside the theorem; the statement of Theorem 11 on p. 16 ends with them, so Part I’s original citation [2, Theorem 11] stands, and the $O(N^4)$, $O(N^3)$ and $O(N^5)$ forms are those of the proof and discussion.) Their conclusion (p. 37) says that “the optimal cost of the computation of pop-stacked permutations is still open”; the recurrence (8.1) lowers the operation count and leaves that question open (Section 33, item 5).

The independent verification route expands (1.2)–(1.4) as rational ordinary power series, divides by D , and multiplies the n th coefficient by $n!$. It does not use the Riccati recurrence. Both routes match all 71 fixture counts $p_0, \dots, p_{70}$ exactly. The fixture records separate provenance: $p_1, \dots, p_{25}$ are the posted OEIS values checked in the earlier source work; the longer fixture was computed independently from the published endpoint recurrence. Agreement is finite evidence, not the proof of the formula.

[write] **6 October 2026.** The “earlier source work” is Report 161 (Part II): the fixture `data/164-egf-d ata-counts_p0_p70.json` names `Report161/data/counts.json` with its SHA-256 `d42dae1b...`, which is the SHA-256 of the shipped `data/161-operator-data-counts.json` (checked at the write). The write also ran (8.1) to $n = 450$: it reproduces all 450 terms of the OEIS b-file of A307030 [9], and $6 \mid$ the right-hand side at every step.

9 Unique dominant pole and leading asymptotic

For completeness we supply the positive-operator isolation argument, rather than infer a complex gap from the positive pole equation. Let $\mathcal{K}$ denote the integral operator with kernel K from (2.1). For real $t > 0$, multiplication by $\sqrt{w_t}$ is boundedly invertible on $L^2(X, \mu)$. Thus A_t is similar to the compact self-adjoint, positivity-preserving operator

$$S_t = M_{\sqrt{w_t}} \mathcal{K} M_{\sqrt{w_t}}.$$

It need not be positive semidefinite. For any states s, s' , every interval (a, b) with

$$0 < a < \min(\text{lo}(s), \text{lo}(s')), \quad \max(\text{hi}(s), \text{hi}(s')) < b < 1$$

overlaps both. This set has positive area. Consequently S_t^2 has a strictly positive kernel almost everywhere.

Lemma 9.1. *For such an operator, $\lambda(t) = \text{spr}(A_t) = \|S_t\| > 0$ is a simple positive eigenvalue with positive left and right eigenfunctions. All other eigenvalues have smaller modulus.*

Proof. The compact self-adjoint spectral theorem gives a top eigenvector of S_t^2 . Replacing it by its absolute value cannot decrease its Rayleigh quotient, so the absolute value is also a top eigenvector. Strict positivity of the squared kernel makes it positive almost everywhere. A sign-changing top eigenvector would yield a strictly larger Rayleigh quotient after taking its absolute value, a

contradiction. Hence the top eigenspace is one-dimensional: two independent real vectors would produce a nonzero vector orthogonal to a positive vector, which must change sign. The action of S_t on the positive vector is positive, so its eigenvalue is $+\|S_t\|$. Any other eigenvalue of the same modulus would give another top vector of S_t^2 . Compactness gives the strict spectral gap. Similarity transfers simplicity and positive left and right eigenvectors to A_t . $\square$

Norm continuity gives continuity of λ . Increasing t strictly increases the kernel of S_t wherever $K = 1$; testing on its positive Perron vector proves strict increase of λ . Also $\lambda(t) \rightarrow 0$ as $t \downarrow 0$. On $F = \{(a, b) : a < 1/4, b > 3/4\}$, all pairs overlap, $\mu(F) = 1/16$, and $w_t \geq t^2 e^{t/2}$. Its indicator Rayleigh quotient gives

$$\lambda(t) \geq t^2 e^{t/2} / 16 \rightarrow \infty.$$

Thus there is a unique $t_* > 0$ with $\lambda(t_*) = 1$.

The simple eigenvalue depends analytically on the parameter locally. If $f_t > 0$ is its real unit eigenvector for S_t , differentiation gives

$$\lambda'(t) = \lambda(t) \int_X \frac{w'_t}{w_t} f_t^2 d\mu > 0. \quad (9.1)$$

One can obtain this directly from $S'_t = \frac{1}{2}(M_{w'_t/w_t} S_t + S_t M_{w'_t/w_t})$ and the normalized simple-eigenvalue derivative formula. Local analyticity follows from a Riesz projection around the isolated eigenvalue, choosing the positive weight square root locally.

At t_* put $h_* = M_{1/\sqrt{w_{t_*}}} f_{t_*}$ and $\ell_* = M_{\sqrt{w_{t_*}}} f_{t_*}$. Their pairing is one. The local resolvent singular part is the corresponding analytic rank-one projection divided by $1 - \lambda(z)$. Its scalar numerator in (2.4) is

$$\left(\int_X \sqrt{w_{t_*}} f_{t_*} d\mu \right)^2 > 0.$$

Equation (9.1) proves that this is a genuine simple positive scalar pole. For $|z| < t_*$, the absolute kernel comparison $|A_z^k f| \leq A_{|z|}^k |f|$ and the spectral-radius formula show $\text{spr}(A_z) \leq \lambda(|z|) < 1$; hence there are no poles inside. The scalar resolvent identity extends over this disk by its invertible analytic resolvent. It follows that $t_* = \rho_0$ from (5.4). Write $\rho = \rho_0$.

For $|z| = \rho$, $z \neq \rho$, define $B_z = \mathcal{K} M_{|w_z|}$. Its weights are positive and bounded away from zero. The same squared-kernel argument gives a positive Perron vector h_z with eigenvalue $\beta = \text{spr}(B_z)$. On the interval component,

$$|w_z(a, b)| = \rho^2 e^{\text{Re}(z)(b-a)} < w_\rho(a, b),$$

whereas the singleton weights have equal absolute value. Using the positive left eigenvector ℓ_* of A_ρ yields

$$(1 - \beta) \int_X \ell_* h_z d\mu = \int_\Delta (w_\rho - |w_z|) h_z (\mathcal{K} \ell_*) d\mu > 0.$$

Indeed $\mathcal{K} \ell_* = \ell_*/w_\rho > 0$ almost everywhere. Thus $\text{spr}(A_z) \leq \beta < 1$ at every other point of this circle. Invertibility near such a point and continuation from the interior identify the quotient with the analytic scalar resolvent there.

We already know from (1.5) that scalar poles form a discrete set. Compactness of the circle and discreteness, including at ρ , give some $R > \rho$ with no other scalar pole on or inside that disk. Subtracting the ρ pole and applying Cauchy's estimate proves

Theorem 9.2. *There exist $R > \rho$ and $0 < \theta = \rho/R < 1$ such that*

$$p_n = C n! \rho^{-n} (1 + O(\theta^n)), \quad C = \frac{2e^{-2\rho}(4e^\rho - 1)}{\rho(\rho + 2)} > 0. \quad (9.2)$$

The positive pole ρ is the unique dominant pole.

The residue formula gives C directly; equivalently $C = e^{-\rho} r(\rho)/(\rho \phi'(\rho))$. This proof establishes existence of a complex pole-free gap, but supplies no certified numerical value of R , error constant, or onset index. The real interval certificate below does not fill that separate gap.

[write] Second routes. Part II proves Theorem 9.2 without the explicit formula (Theorem 14.1, Sections 16–18), writing out the analytic Fredholm continuation that is used here through the explicit quotient. The constants agree: Part II’s ρ is the unique crossing $\lambda(\rho) = 1$, which is $t_* = \rho_0$ above, and Part II’s C of (14.6) is the amplitude of the principal part $C/(1 - z/\rho)$ of the same function P at the same simple pole (17.3), hence equal to $-\text{Res}(P, \rho)/\rho$, the closed form in (9.2). Lemma 9.1 is Part II’s Lemma 16.1 and Part III’s Lemma 25.1; Part III, Section 25, repeats this isolation at $u = 1$.

10 Rational certificates for the constants

Let the terminating decimals below denote exact rationals:

$$\begin{aligned} L &= 1.11343904173672704376166152691808324014139016583344946615, \\ U &= 1.11343904173672704376166152691808324014139016583344946616, \\ C_L &= 0.695688549070635767995703168724110156574198350721, \\ C_U &= 0.695688549070635767995703168724110156574198350722. \end{aligned}$$

The companion proves with integer and rational arithmetic that

$$L < \rho < U, \quad C_L < C < C_U. \quad (10.1)$$

The following remainder bounds make the certificate reproducible without assuming the correctness of numerical transcendental functions.

For $0 \leq x < 202$, let $T_{200}(x) = \sum_{j=0}^{200} x^j/j!$. Each ratio between consecutive terms in the tail after $x^{201}/201!$ is at most $x/202$. Hence

$$T_{200}(x) \leq e^x \leq T_{200}(x) + \frac{x^{201}/201!}{1 - x/202}. \quad (10.2)$$

Monotonicity of e^x extends these bounds to interval inputs. For a nonnegative rational x , put $S = 10^{120}$ and $k = \lfloor \sqrt{\lfloor xS^2 \rfloor} \rfloor$ using integer square root. Checked integer squares prove

$$(k/S)^2 \leq x < ((k+1)/S)^2,$$

which encloses $\sqrt{x}$. For $0 \leq x \leq 1$, the sine and cosine Taylor terms decrease in magnitude. Truncating each alternating series at index 60 gives an upper bound, because the last included term is positive; subtracting the next term’s magnitude gives a lower bound. Sine is increasing and cosine decreasing on this interval, so their endpoint bounds propagate interval inputs rigorously.

For $x = L, U$ the program evaluates

$$(2e^{x/2} + 1) \cos\left(\frac{x\sqrt{4e^x - 1}}{4}\right) - \sqrt{4e^x - 1} \sin\left(\frac{x\sqrt{4e^x - 1}}{4}\right)$$

by those rational interval operations. It verifies that both sine/cosine arguments are in $(0, 1)$ and that $D(L) > 0 > D(U)$. In this bracket $0 < zr/4 < 1$ and $0 < \alpha < \pi/4$, so $\phi < 1 + \pi/4 < \pi$. Only the first positive tangent pole can be crossed. Continuity and strict increase from (5.3) therefore isolate ρ_0 . Finally direct positive interval substitution into

$$C = \frac{2(4e^\rho - 1)}{e^{2\rho}\rho(\rho + 2)}$$

proves the amplitude bracket. The machine-readable certificate records outward decimal enclosures of both denominator signs and the propagated amplitude interval. Decimal endpoints are suggested values only until these rational inequalities have been verified; no binary floating point or fitted coefficient is trusted by the certificate.

[write] **6 October 2026.** Part III's certificate (28.1) uses the same endpoints L, U , digit for digit, and adds $U < 6/5 < \log 4$. The write reran the companion's certificate on a fresh extraction; its output reproduced `data/164-egf-data-constant_certificate.json`. The interval check of Remark 5.1 proves all 75 decimals of ρ printed by Claesson, Guðmundsson and Pantone, more than the 56 certified here; their 58 printed decimals of C agree with a 220-digit evaluation of the closed form (certified here to 48 decimals only).

11 Stirling expansions and safe inversion

For sufficiently large integers, Theorem 9.2 gives

$$\log p_n = \mathcal{H}(n) + \epsilon_n, \quad \mathcal{H}(x) = \log \Gamma(x + 1) - x \log \rho + \log C, \quad |\epsilon_n| \leq M\theta^n \quad (11.1)$$

for some fixed $M > 0$, $0 < \theta < 1$ and onset n_0 . For each fixed $m \geq 1$, the ordinary Stirling remainder then gives

$$\begin{aligned} \log p_n &= n \log n - (1 + \log \rho)n + \frac{1}{2} \log n + \log(C\sqrt{2\pi}) \\ &\quad + \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)n^{2j-1}} + O(n^{-2m-1}). \end{aligned} \quad (11.2)$$

Exponentiating a sufficiently long finite truncation gives, for every fixed $K \geq 0$,

$$p_n = C\sqrt{2\pi n} \left(\frac{n}{e\rho}\right)^n \left(\sum_{k=0}^K \frac{s_k}{n^k} + O(n^{-K-1})\right),$$

where s_k are the formal coefficients of

$$\exp\left(\sum_{j \geq 1} \frac{B_{2j}t^{2j-1}}{2j(2j-1)}\right) = 1 + \frac{t}{12} + \frac{t^2}{288} - \frac{139t^3}{51840} + \cdots.$$

These are fixed-order Poincaré expansions; no convergence of the infinite Bernoulli series is claimed. Also $p_{n+1}/p_n = ((n+1)/\rho)(1 + O(\theta^n))$.

Define $N(y) = \min\{n \geq 0 : p_n \geq y\}$ for $y > 0$. Appending the new maximum extends the last ascending run and preserves the overlap inequalities, giving an injection from size n to size $n + 1$. Thus (p_n) is nondecreasing; the ratio formula makes it eventually strictly increasing and unbounded. For large y , let x be the unique large solution of $\mathcal{H}(x) = \log y$. This is the eventually increasing branch because $\mathcal{H}'(x) = \psi(x + 1) - \log \rho \sim \log x$ and $\mathcal{H}''(x) = \psi'(x + 1) > 0$.

Proposition 11.1 (Rounding-safe threshold). *If (11.1) holds for all $n \geq n_0$ and*

$$x - 2 \geq n_0, \quad \mathcal{H}'(x - 2) > 0, \quad \delta(x) = \frac{2M\theta^{x-2}}{\mathcal{H}'(x - 2)} < 1,$$

then

$$\lceil x - \delta(x) \rceil \leq N(y) \leq \lceil x + \delta(x) \rceil. \quad (11.3)$$

Proof. Set $a = \lceil x - \delta \rceil - 1$ and $b = \lceil x + \delta \rceil$. Both lie in $[x - 2, x + 2]$. The increasing positive derivative of $\mathcal{H}$ gives $\mathcal{H}(x) - \mathcal{H}(a) > 2M\theta^{x-2}$ and $\mathcal{H}(b) - \mathcal{H}(x) \geq 2M\theta^{x-2}$. The absolute errors at a, b are at most $M\theta^{x-2}$. Hence $p_a < y \leq p_b$, and monotonicity gives the bracket. $\square$

If the two ceilings agree, they determine the exact threshold; otherwise exact counting or stronger information is required. In particular one cannot automatically round an asymptotic root, even if its error tends to zero. Here $\delta(x) = O(\theta^x / \log x)$. The certified ρ, C intervals alone do not provide effective M, θ, n_0 for this implication.

For a convenient real starting estimate, write $L_y = \log y$ and $u = W(L_y/(e\rho))$ on the positive real Lambert branch, so $x_0 = L_y/u$ satisfies $x_0 \log(x_0/(e\rho)) = L_y$. One Newton correction in the Stirling expansion gives

$$x = x_0 - \frac{\frac{1}{2} \log x_0 + \log(C\sqrt{2\pi})}{u + 1} + O\left(\frac{1}{x_0 u}\right). \quad (11.4)$$

Indeed the displayed correction is $O(1)$, the omitted residual is $O(1/x_0)$, and the derivative is asymptotic to $u + 1$. This continuous inverse estimate is not an exact integer-ceiling rule.

[write] Part II, Section 19, proves the same Stirling expansions (Corollary 19.1), the same bracket with the same proof (Theorem 19.2), the same first inverse correction (19.8), and every fixed order of the continuous inverse (Proposition 19.3). Its $u = \log(x_0/\rho)$ is the present u plus one, so its denominator $\log(x_0/\rho)$ is the present $u + 1$, and its numerator $\frac{1}{2} \log(2\pi x_0) + \log C$ equals the present one. The front matter (“The inverses and the transseries volume”) classifies x_0 , (11.4) and Proposition 11.1.

12 Source status and claim boundaries

The underlying overlap characterization is due to Asinowski, Banderier, Billey, Hackl and Linusson [1], with the later development by Asinowski, Banderier and Hackl [2]. Claesson, Guðmundsson and Pantone [3] give polynomial-time counting and extensive numerical experiments. Their author PDF, page 2, leaves a total-count generating function or closed enumeration formula open. Section 3 distinguishes unsuccessful bounded symbolic fits from experimental differential approximants. In particular those fits do not prove non-D-finiteness or differential transcendence. Their Section 3.3 predicts positive and nonreal simple poles and the leading asymptotic numerically.

The authors’ linked repository [12] already contains estimates of ρ, C , positive poles, and nonreal poles. For orientation, its first positive estimates begin 1.113439..., 2.417184..., 3.076673...; these

decimal observations are prior art. Neither their numerical discovery nor a claimed numerical complex-pole ordering is a contribution of this report. The new statements here are the exact formula, equations, residues, rational bounds, and proofs.

The 2021 paper [2], Section 2.4, records the exact asymptotic problem and the infinitely-many-singularities/non-D-finiteness question as unresolved. Its Theorem 10 is a multivariate ordinary functional equation, distinct from the present one-variable explicit EGF. Theorem 11 gives $O(N^4)$ arithmetic operations and $O(N^3)$ simultaneous scalar allocations for total counts. The counting paper [3], Section 2, gives the same operation and memory orders and explicitly separates constant-cost arithmetic from its multiprecision implementation. Neither comparison is a bit-complexity claim.

The final published 2023 version of Defant and Williams’ *Semidistrim Lattices* [5] was also checked, not only its 2021 preprint. Its Section 11.2 and Figure 12 connect the type-A image count to A307030, while Remark 11.3 records later progress on neighboring problems. The cited subsequent Choi–Sun work [4] concerns other Coxeter/Tamari/root-poset settings and does not provide the present classical type-A total-count EGF. The current OEIS source entry [9] was read as revision 63 dated 18 November 2021; its lack of a formula field corroborates but cannot establish novelty.

A bounded source search on 3 October 2026 covered these primary papers, the authors’ live repository, the current OEIS entry, and exact or near-exact searches combining pop-stacked permutations or A307030 with generating function, closed form, exponential, D-finite, Riccati, differential equation, and quadratic enumeration. No matching explicit total-count EGF, proved non-D-finiteness, exact nonlinear ODE, or quadratic arithmetic total-count recurrence was located. This is a bounded search result, not proof of worldwide priority; scholarly author and citation review remains appropriate before any submission. No publication, author contact, or repository edit was undertaken.

[write] **6 October 2026.** What the write re-read (front matter, “The predictions of Claesson, Guðmundsson and Pantone”). The authors’ repository [12] lists twenty positive and eight nonreal estimates and the constant C ; its first positive estimates are as quoted above, and it lists the pole ρ_{12} that the paper’s printed list omits (Remark 5.1). Asinowski–Banderier–Hackl [2, Section 2.4] call an exact asymptotic analysis “a challenge” and report the conjectures of [3] (p. 18); their Theorem 10 is a functional equation for a five-variable generating function. The OEIS entry [9] is still revision #63. Defant–Williams [5] and Choi–Sun [4] were not re-read by the write.

13 Companion and reproducible release

The source package contains a standard-library Python companion with three bounded public modes: count through at most $N = 400$, independently verify quotient coefficients through at most $N = 70$, and certify the fixed real constants. It rejects invalid types, negative indices and out-of-range requests before allocating a large computation. The public computational API performs no writes; command-line JSON goes to standard output, and errors go to standard error. Mathematical gates use explicit exceptions and remain active under Python’s optimization flag.

The quotient check is intentionally a separate, slower exact series calculation. It is not covered by the recurrence’s quadratic arithmetic bound. The certificate uses exact fractions, Taylor bounds and integer square roots as proved in Section 10. The data fixture and output receipts distinguish source values from computed values.

PDF reproduction uses a fresh exclusive output directory, disabled TeX shell escape, fixed metadata, a bounded stabilization loop, and settled reference/layout checks. ZIP reproduction uses

a fixed allowlist, SHA-256 verification, fixed member metadata, and exclusive no-clobber output. The I/O helpers reject symbolic-link path components and pin opened parent directories. This assumes trusted source files and trusted output parent directories; it is not a sandbox for malicious TeX or a hostile user who can replace files inside those directories.

Normal and optimized tests, two clean PDF builds, two ZIP creations, and all-page rendered inspection are release checks. A deterministic archive packages only the allowlisted report, source, fixtures, certificates and reproduction scripts; temporary logs and page images are excluded. Consult the package README for the exact commands and scope of the final receipts.

[write] In this directory the companion is `code/164-egf-companion-pop_egf.py` with its tests and README (`164-egf-companion-README.md`), the data are `data/164-egf-data-*.json`, and the release tools are `code/164-egf-*.py`. The package README is replaced by the report's README; the PDF and SHA256SUMS are not shipped. Nothing runs in this directory under the shipped names; the README gives the rerun route from the arrival commit.

Part II

Factorial exponential growth of pop stacked permutations: an operator proof and asymptotic inversion

Bundle Report 161, archive `Pop_Stacked_Permutations_Asymptotics_and_Inverses_Source.zip`, delivered as `Report161.tex` (13 pages). Delivered title block “Factorial exponential growth of pop stacked permutations / An operator proof and asymptotic inversion / Report 161 / 3 October 2026”. Printed as delivered, with `[write]` notes; Report 161’s Section k is Section $k + 13$ here, and its statements and equations move with their sections. The opening paragraph “Attribution and scope” is the manuscript’s own.

Abstract of this Part (as delivered). Let p_n be the number of permutations of $[n]$ in the image of one pass through a pop stack, equivalently the number whose adjacent maximal ascending runs overlap. This is OEIS A307030. We prove

$$p_n = C n! \rho^{-n} (1 + O(q^n)), \quad \rho > 1, \quad C > 0, \quad 0 < q < 1.$$

The constants are specified exactly by a compact integral operator on a disjoint union of singleton and interval run states. The proof includes the labelled-simplex normalization, a self-adjoint Perron argument, an analytic Fredholm reduction, a positive simple pole, and strict domination at every other point on its circle of convergence. It yields all fixed orders of the Stirling expansion, a recursively specified all-orders expansion for the continuous inverse, and an exponentially narrow ceiling bracket around the exact log-Gamma root.

The factorial-exponential law was predicted numerically in the counting literature. The spectral method itself is classical; the contribution here is its variable-run construction and the resulting proof for this sequence. Published decimal estimates of ρ and C are not certified by this report. Non-D-finiteness, further poles, and an effective finite-input asymptotic threshold certificate are not proved. A standard-library companion supplies independently structured finite checks and exact bounded threshold searches.

Attribution and scope. The overlap characterization and exact counting algorithms are prior results of Asinowski and collaborators [1, 2] and Claesson, Guðmundsson and Pantone [3]. Positive integral operators and spectral asymptotics for permutation classes were developed by Ehrenborg, Kitaev and Perry [6]. The new claim is the theorem proved below for the total pop-stacked count, not invention of those methods. A bounded check of the current OEIS entry, the primary pop-stack papers, *Semidistributive Lattices*, and the available report catalogue found no earlier proof of this leading equivalent. This is not exhaustive scholarly priority clearance. Section 20 records exactly what the source gate and the finite evidence support.

Reading conventions of Part II [write]

p_n, P	as in Part I
$J, D, \mu, X = I \sqcup D$	overlap kernel and interval states (Part I's K and Δ), measure, states
$S_r, \lambda, v, d_\rho, h, \ell$	symmetrized operator, Perron eigenvalue, eigenfunction, $\lambda'(\rho)$, right and left eigenvectors
ρ, C, q, R	Part I's ρ and C (note in Section 14); $q = \rho/R$ (Part I's θ)
H, x, x_0, u, b	log-Gamma model (Part I's $\mathcal{H}$), its root, the initializer, $u = \log(x_0/\rho)$ (Part I's u plus one), $b = \frac{1}{2} \log(2\pi x_0) + \log C$
$d_m, \mathcal{F}$	inverse coefficients, the formal series (19.9) (not Part I's resultant)
$F, E_z, B_z, D(z)$	the Fredholm reduction of Section 17 (this D is a matrix, not Part I's denominator)

14 Definitions and main theorem

A pop-stack pass reverses each maximal decreasing run of a permutation and concatenates the resulting increasing blocks. A permutation in its image is called *pop-stacked*. Put $p_0 = 1$ for the empty permutation and let p_n count pop-stacked permutations of $[n]$ for $n \geq 1$. The first values, with the empty term included, are

$$1, 1, 1, 3, 11, 49, 263, 1653, 11877, 95991, 862047.$$

All runs below are maximal ascending runs unless stated otherwise. For consecutive runs with endpoints (a, b) and (c, d) the two required strict inequalities are

$$a < d, \quad c < b. \quad (14.1)$$

The second says that the boundary is a descent; the first is the additional image condition. A singleton has coincident endpoints. Thus the terminology “overlap” means these cross inequalities, not a positive-length intersection: a singleton may lie strictly inside the range of a neighboring nonsingleton run. Two singleton runs can never satisfy both inequalities.

Here is an exact specification of the constants. Let

$$I = (0, 1), \quad D = \{(a, b) : 0 < a < b < 1\}, \quad X = I \sqcup D, \quad \mu = dx \sqcup da db,$$

so the Hilbert space is $\mathcal{H} = L^2(X, \mu) = L^2(I) \oplus L^2(D)$. For $s = x \in I$ set $\text{lo}(s) = \text{hi}(s) = x$; for $s = (a, b) \in D$ set $\text{lo}(s) = a$, $\text{hi}(s) = b$. Define the symmetric kernel

$$J(s, t) = \mathbf{1}\{\text{lo}(s) < \text{hi}(t), \text{lo}(t) < \text{hi}(s)\}, \quad (Jf)(s) = \int_X J(s, t)f(t) d\mu(t), \quad (14.2)$$

and the entire weights

$$w_z(x) = z \quad (x \in I), \quad w_z(a, b) = z^2 e^{z(b-a)} \quad ((a, b) \in D). \quad (14.3)$$

Write M_f for multiplication by f . For $r > 0$ put

$$S_r = M_{\sqrt{w_r}} J M_{\sqrt{w_r}}, \quad \lambda(r) = \|S_r\|.$$

As proved below, $\lambda(r)$ is the simple positive leading eigenvalue, strictly increases from zero to infinity, and has a unique crossing

$$\lambda(\rho) = 1. \quad (14.4)$$

Let $v = v_\rho > 0$ be its real eigenfunction normalized by $\|v\|_2 = 1$. Define

$$d_\rho = \frac{1}{\rho} \int_I v(x)^2 dx + \int_D \left(\frac{2}{\rho} + b - a \right) v(a, b)^2 da db, \quad (14.5)$$

$$C = \frac{\left(\int_X \sqrt{w_\rho(s)} v(s) d\mu(s) \right)^2}{\rho d_\rho}. \quad (14.6)$$

There is no fitted or numerical parameter in these definitions.

Theorem 14.1 (Factorial-exponential equivalent). *The definitions above give unique constants ρ and C , with*

$$(24/23)^{1/4} \leq \rho \leq 2, \quad C > 0. \quad (14.7)$$

For some $R > \rho$,

$$\frac{p_n}{n!} = C\rho^{-n} + O(R^{-n}). \quad (14.8)$$

Equivalently, with $q = \rho/R \in (0, 1)$, $p_n = Cn!\rho^{-n}(1 + O(q^n))$. The exponential generating function $P(z) = \sum_{n \geq 0} p_n z^n / n!$ continues meromorphically to $\mathbb{C}$. Its only singularity in some closed disk of radius greater than ρ is the simple pole at ρ , with principal part $C/(1 - z/\rho)$.

The scalar function may have cancellations of other operator poles. No assertion is made about their number, positions, or existence. In particular, meromorphic continuation and one pole do not imply non-D-finiteness. The published predictions $\rho \approx 1.113439041736727\dots$ and $C \approx 0.695688549070635\dots$ [3, Section 3.3] are useful context only. We do not certify any of their decimal places or identify a numerical isolation interval narrower than (14.7).

[write] **6 October 2026.** Superseded, and kept as delivered. Part I, written after this Part on the same day, proves that P is meromorphic with exactly the poles $\{D = 0, Q \neq 0\}$, all simple, with infinitely many positive ones, and that P is not D-finite (Sections 5–6); it certifies ρ to 56 and C to 48 decimals (10.1); and the write’s interval check proves all 75 printed decimals of ρ (Remark 5.1). The constants here are Part I’s: the crossing $\lambda(\rho) = 1$ is Part I’s $t_* = \rho_0$, and C of (14.6), the amplitude of the principal part (17.3), equals $-\text{Res}(P, \rho)/\rho = 2e^{-2\rho}(4e^\rho - 1)/(\rho(\rho + 2))$ by Part I’s (1.8). So Theorem 14.1 is a second route to Part I’s Theorem 9.2.

15 The labelled run integral

Lemma 15.1 (Image characterization). *A permutation is pop-stacked if and only if every consecutive pair of its maximal ascending runs satisfies (14.1).*

Proof. For necessity, write an input permutation as its maximal decreasing blocks, and let $B_1, B_2, \dots$ be their reversals. The boundary between two original blocks is an ascent, so $\min B_i < \max B_{i+1}$. Each output block is increasing. The maximal ascending runs of the output are obtained by merging successive blocks at ascent boundaries. Between two consecutive merged groups, let B_i be the last block of the left group and B_{i+1} the first block of the right group. Since the merged groups are increasing,

$$\min(\text{left}) \leq \min B_i < \max B_{i+1} \leq \max(\text{right}).$$

The other cross inequality holds because the boundary of maximal ascending runs is a descent.

Conversely, reverse each maximal ascending run of a permutation satisfying (14.1). Each reversed block is decreasing, and the boundary between reversed blocks is an ascent because

$\min(\text{left}) < \max(\text{right})$. They are therefore the maximal decreasing runs of the resulting input permutation. A pop-stack pass reverses them back. This proves the characterization, which is the established result in [1, Theorem 1] and [2, Theorem 3]. $\square$

Fix a composition $n = \ell_1 + \cdots + \ell_k$ into positive run lengths. Place one real coordinate at each of the n actual permutation positions, independently in $(0, 1)$, with product Lebesgue measure. Impose increasing inequalities within the proposed blocks and both cross inequalities at every adjacent block boundary. Away from the null sets where coordinates tie, the rank order of the coordinates is a permutation. Each of the $n!$ labelled rank-order simplexes has volume exactly $1/n!$; this follows by coordinate permutation from the disjoint partition of the unit cube into these simplexes. Thus the constrained volume is exactly the number of admissible permutations with this run composition divided by $n!$.

There is no further multinomial coefficient or run factorial. The coordinates belong to actual positions. Also there is no multiple counting over compositions: the second cross inequality makes every proposed boundary a descent, so the composition is exactly the unique maximal-ascending-run composition.

For a block of size $\ell \geq 2$, retain its first and last coordinates $a < b$. Its $\ell - 2$ interior coordinates are increasing between them, so their integrated volume is

$$\frac{(b - a)^{\ell-2}}{(\ell - 2)!}.$$

After multiplying by z^ℓ and summing over all such lengths this is

$$\sum_{\ell \geq 2} \frac{z^\ell (b - a)^{\ell-2}}{(\ell - 2)!} = z^2 e^{z(b-a)}.$$

A singleton instead contributes $z dx$ on its separate one-dimensional component. It is not the zero-measure diagonal of D . For a useful check, the contribution of all single-run objects is

$$\int_X w_z d\mu = z + z^2 \int_0^1 (1 - t) e^{zt} dt = e^z - 1, \quad (15.1)$$

whose coefficient is $1/n!$, as required for the unique increasing permutation of every positive size.

Put $A_z = JM_{w_z}$. The contribution from $k \geq 1$ runs is

$$\int_{X^k} \prod_{i=1}^k w_z(s_i) \prod_{i=1}^{k-1} J(s_i, s_{i+1}) d\mu^k = \int_X w_z A_z^{k-1} \mathbf{1} d\mu.$$

Since $\|A_z\| \rightarrow 0$ as $z \rightarrow 0$, the Neumann series converges in operator norm near zero. The sum of absolute values of these integrals is finite there as well, by the same bound with $|w_z|$. The simplex argument therefore proves the analytic identity

$$P(z) = 1 + \int_X w_z(s) [(I - A_z)^{-1} \mathbf{1}](s) d\mu(s) \quad (|z| \text{ small}). \quad (15.2)$$

The scalar pairing here is bilinear, without complex conjugation; it preserves holomorphic dependence on z .

[write] Part I, Section 2, and Part III, Section 22, prove the same normalization (Part I writes K for J and Δ for D ; Part III adds the marker u). Lemma 15.1 is the characterization of [1, Theorem 1] and [2, Theorem 3], proved here; Parts I and III cite it.

16 Compactness and the Perron crossing

Since $\mu(X) = 3/2$ and J is a measurable bounded kernel, J is Hilbert–Schmidt, hence compact. The map $z \mapsto w_z$ is entire in $L^\infty(X)$: on every compact z -disk the exponential series and all its derivatives converge uniformly because $0 < b - a < 1$. Consequently A_z is an entire family of compact operators, and $A_0 = 0$.

For every fixed $r > 0$, w_r is bounded above and away from zero on both components of X . Thus $T_r = M_{\sqrt{w_r}}$ is boundedly invertible and

$$T_r A_r T_r^{-1} = S_r = T_r J T_r. \quad (16.1)$$

The operator S_r is compact, self-adjoint, and positivity preserving. We do not assert that S_r is positive semidefinite.

For any states s, t set $m = \min(\text{lo}(s), \text{lo}(t)) > 0$ and $M = \max(\text{hi}(s), \text{hi}(t)) < 1$. Every interval state $u = (a, b)$ with $0 < a < m$ and $M < b < 1$ satisfies $J(s, u)J(u, t) = 1$. This is a set of area $m(1 - M) > 0$. It follows that J^2 has a strictly positive integral kernel; the kernel of S_r^2 is likewise strictly positive almost everywhere, as all intervening weights are positive.

Lemma 16.1 (The needed Perron conclusion). *Suppose S is a nonzero compact self-adjoint operator with a nonnegative symmetric integral kernel, and S^2 has a strictly positive kernel almost everywhere. Then $\lambda = \|S\| > 0$ is a simple positive eigenvalue with a real eigenfunction $v > 0$ almost everywhere. Every other eigenvalue has modulus strictly smaller than λ .*

Proof. By the compact self-adjoint spectral theorem, the positive-semidefinite operator S^2 has a real top eigenfunction f with eigenvalue λ^2 . Its nonnegative kernel gives

$$\langle |f|, S^2 |f| \rangle \geq \langle f, S^2 f \rangle = \lambda^2 \|f\|_2^2.$$

The norm bound forces equality, and the spectral theorem then shows that $|f|$ is a top eigenfunction. Applying the strictly positive kernel shows $|f| > 0$ almost everywhere. If f had positive and negative sets both of positive measure, the displayed inequality would be strict, by integrating the strictly positive kernel across those sets. Hence every nonzero real vector in this top eigenspace has constant sign. The eigenspace is one-dimensional: two independent real vectors would give a nonzero one orthogonal to a positive vector, which must change sign. Since S preserves this eigenspace and positivity, its action on the positive unit vector is $+\lambda$, not $-\lambda$. Any eigenvalue of modulus λ would give another vector in the top eigenspace of S^2 . Finally compactness excludes accumulation of nonzero eigenvalues at λ or $-\lambda$, giving the strict gap. $\square$

Apply the lemma to S_r . Similarity transfers algebraic simplicity to A_r and gives positive right and left eigenfunctions. In particular $\lambda(r) = \text{spr}(A_r) = \|S_r\|$. Norm continuity gives continuity of λ . If $r_2 > r_1 > 0$, the kernel of $S_{r_2} - S_{r_1}$ is positive wherever $J = 1$. Testing it against the positive unit Perron vector at r_1 and using the Rayleigh bound gives

$$\lambda(r_2) \geq \langle v_{r_1}, S_{r_2} v_{r_1} \rangle > \lambda(r_1).$$

Also $\lambda(r) \rightarrow 0$ as $r \downarrow 0$. On $E = \{(a, b) : 0 < a < 1/4, 3/4 < b < 1\}$ all pairs satisfy $J = 1$, $\mu(E) = 1/16$, and $w_r \geq r^2 e^{r/2}$. Testing against $\mathbf{1}_E / \sqrt{\mu(E)}$ gives

$$\lambda(r) \geq r^2 e^{r/2} / 16 \longrightarrow \infty. \quad (16.2)$$

This proves the existence and uniqueness in (14.4).

The simple eigenvalue is analytic locally near any positive r . For completeness, choose the analytic square root of z near r on I , and $ze^{z(b-a)/2}$ on D . This gives a locally analytic S_z . A small contour around the isolated simple eigenvalue defines an analytic rank-one Riesz projection by its resolvent; its rank stays one, and the restricted operator gives the analytic eigenvalue. For real r , put $D_r = M_{w'_r/w_r}$. Differentiation yields

$$S'_r = \frac{1}{2}(D_r S_r + S_r D_r), \quad \lambda'(r) = \lambda(r) \int_X \frac{w'_r}{w_r} v_r^2 d\mu > 0.$$

At $r = \rho$ this is $\lambda'(\rho) = d_\rho$, exactly (14.5). This proves a simple, transverse crossing.

[write] Lemma 16.1 is stated here for a general operator; Part I's Lemma 9.1 and Part III's Lemma 25.1 are the same argument for S_t .

17 Meromorphic continuation and the positive pole

We give the analytic Fredholm argument used here rather than merely invoking its conclusion. Fix z_0 . Compactness allows a finite-rank $F = UV$, with $U : \mathbb{C}^m \rightarrow \mathcal{H}$ and $V : \mathcal{H} \rightarrow \mathbb{C}^m$, such that $\|A_{z_0} - F\| < 1$. In a small disk around z_0 set $E_z = A_z - F$; there $B_z = I - E_z$ has an analytic Neumann inverse. Then

$$I - A_z = B_z(I - B_z^{-1}UV).$$

The second factor is invertible exactly when the finite-dimensional matrix $D(z) = I - VB_z^{-1}U$ is invertible. Where it is,

$$(I - A_z)^{-1} = B_z^{-1} + B_z^{-1}UD(z)^{-1}VB_z^{-1}. \quad (17.1)$$

The determinant of D is analytic. Either it vanishes identically in this disk, or its zeros are isolated and (17.1) is meromorphic there with finite-order poles. At zero the inverse exists because $A_0 = 0$. Continue along a path from zero to any specified point using a finite chain of overlapping disks of the kind just described. The overlap with a previously reached disk contains invertible points outside its isolated exceptional set; therefore the determinant in the next disk cannot vanish identically. This excludes the first alternative throughout the connected plane. It proves that the noninvertible set is locally discrete and the operator inverse is meromorphic on $\mathbb{C}$.

The functional $f \mapsto \int_X w_z f d\mu$ is entire in operator norm, and $\mathbf{1} \in \mathcal{H}$, so (15.2) continues P meromorphically. At the crossing, put

$$h = T_\rho^{-1}v, \quad \ell = T_\rho v, \quad \int_X \ell h d\mu = 1.$$

These are positive right and left eigenvectors of A_ρ . The rank-one spectral projection is $f \mapsto h \int_X \ell f d\mu$. Locally the singular resolvent part is this analytic projection divided by $1 - \lambda(z)$. Its scalar numerator at ρ is

$$\left(\int_X w_\rho h d\mu \right) \left(\int_X \ell d\mu \right) = \left(\int_X \sqrt{w_\rho} v d\mu \right)^2 > 0. \quad (17.2)$$

Since

$$1 - \lambda(z) = \rho \lambda'(\rho)(1 - z/\rho) + O((z - \rho)^2),$$

we obtain

$$P(z) = \frac{C}{1 - z/\rho} + \text{an analytic germ at } \rho, \quad (17.3)$$

with exactly the positive amplitude (14.6). In particular the scalar pole does not cancel.

18 Strict complex domination and coefficient extraction

For $|z| = r > 0$ define $Q_z = JM_{|w_z|}$. On I , $|w_z| = r$; on D ,

$$|w_z(a, b)| = r^2 e^{\operatorname{Re}(z)(b-a)} \leq w_r(a, b). \quad (18.1)$$

For every $f \in \mathcal{H}$ and every positive integer k ,

$$|A_z^k f| \leq Q_z^k |f| \leq A_r^k |f|.$$

Norms and the spectral-radius formula give

$$\operatorname{spr}(A_z) \leq \operatorname{spr}(Q_z) \leq \lambda(r). \quad (18.2)$$

Thus $I - A_z$ is invertible for $|z| < \rho$.

Suppose now that $|z| = \rho$ and $z \neq \rho$. Then $\operatorname{Re}(z) < \rho$, and (18.1) is strict on the entire interval component D . The positive weight $|w_z|$ is bounded above and away from zero. Applying the previous similarity and squared-kernel proof to Q_z , let $Q_z h_z = \beta h_z$ with $h_z > 0$ and $\beta = \operatorname{spr}(Q_z) > 0$. Use the positive left eigenvector ℓ of A_ρ above. All quantities in the following identity are real:

$$\begin{aligned} (1 - \beta) \int_X \ell h_z \, d\mu &= \int_X \ell (A_\rho - Q_z) h_z \, d\mu \\ &= \int_D (w_\rho - |w_z|) h_z (J\ell) \, d\mu > 0. \end{aligned} \quad (18.3)$$

Indeed $A_\rho^* \ell = w_\rho J\ell = \ell$, so $J\ell = \ell/w_\rho > 0$ almost everywhere. Thus $\beta < 1$, and (18.2) proves invertibility at every other point of the circle. This establishes strictness despite the unchanged absolute singleton weight; no unproved aperiodicity assertion is needed.

The noninvertible set is locally discrete and has no point in $|z| \leq \rho$ except ρ . Consequently there is an $R > \rho$ with no other exceptional point on or inside $|z| \leq R$. To see the uniformity explicitly, otherwise exceptional points with moduli decreasing to ρ would have a convergent subsequence to a point of the compact circle; that would contradict local discreteness, including at ρ . Shrink R if necessary so that an open neighborhood of its closed disk has the same property. By (17.3),

$$G(z) = P(z) - \frac{C}{1 - z/\rho}$$

is analytic on that neighborhood. Cauchy's coefficient bound gives $[z^n]G = O(R^{-n})$, which proves (14.8).

It remains to prove (14.7). Four consecutive decreasing entries $a > b > c > d$ force the two middle entries to be consecutive singleton ascending runs. This is forbidden by (14.1). For a uniform random permutation, the relative orders in disjoint positional blocks of four are independent and uniform, as follows also from independent continuous coordinates. Hence

$$\frac{p_n}{n!} \leq (23/24)^{\lfloor n/4 \rfloor}. \quad (18.4)$$

Taking n th roots and using (14.8) gives $\rho \geq (24/23)^{1/4} > 1$. Four entries, rather than three, are essential: 14325 is pop-stacked and contains the decreasing triple 432.

For the upper bound, interleave any ordering of the lower m values with any ordering of the upper m values, beginning with a lower value. The maximal ascending runs are the m low-high

pairs, and all neighboring pairs satisfy (14.1). Thus $p_{2m} \geq (m!)^2$ and $p_{2m}/(2m)! \geq \binom{2m}{m}^{-1}$. Taking $2m$ th roots gives $\rho \leq 2$. Theorem 14.1 is proved.

[write] **6 October 2026.** The interleaving lower bound $p_{2m} \geq (m!)^2$ is the construction of Asinowski, Banderier and Hackl [2, Theorem 12, Section 2.4] (intertwining two permutations of size $n/2$), which they state as growth at least $\exp(n \ln n - n \ln 2 - n + o(n))$; they also mention a comparison with alternating permutations that gives the lower bound $.63^n n!$. This Part does not cite them. The bracket (14.7), $1.0107 \dots \leq \rho \leq 2$, is true and contains Part I's certified $\rho = 1.11343904 \dots$; it is kept as an elementary bound.

19 All algebraic orders and inverse thresholds

The exponentially small relative error implies, for sufficiently large integers n ,

$$\log p_n = H(n) + \epsilon_n, \quad H(x) = \log \Gamma(x+1) - x \log \rho + \log C, \quad |\epsilon_n| \leq Mq^n, \quad (19.1)$$

for some fixed $M > 0$ and $q \in (0, 1)$. The constants M , q , and an onset n_0 exist; numerical certified values are not supplied.

Corollary 19.1 (Complete fixed-order Stirling expansions). *For every fixed integer $m \geq 1$,*

$$\begin{aligned} \log p_n &= n \log n - (1 + \log \rho)n + \frac{1}{2} \log n + \log(C\sqrt{2\pi}) \\ &\quad + \sum_{j=1}^m \frac{B_{2j}}{2j(2j-1)n^{2j-1}} + O(n^{-2m-1}). \end{aligned} \quad (19.2)$$

Here B_{2j} are the Bernoulli numbers. For every fixed $K \geq 0$,

$$p_n = C\sqrt{2\pi n} \left(\frac{n}{e\rho}\right)^n \left(\sum_{k=0}^K \frac{s_k}{n^k} + O(n^{-K-1})\right), \quad (19.3)$$

where $s_0 = 1$ and the s_k are the formal coefficients of

$$\exp\left(\sum_{j \geq 1} \frac{B_{2j} t^{2j-1}}{2j(2j-1)}\right) = 1 + \frac{t}{12} + \frac{t^2}{288} - \frac{139t^3}{51840} + \dots \quad (19.4)$$

Also $p_{n+1}/p_n = ((n+1)/\rho)(1 + O(q^n))$.

Proof. Insert the ordinary finite-order Stirling expansion of $\log \Gamma(n+1)$ into (19.1); the exponential error is smaller than every fixed algebraic order. Exponentiating a sufficiently long finite truncation gives (19.3). Dividing two successive instances of the leading equivalent gives the ratio. These are Poincaré expansions at fixed order, not claims that an infinite Bernoulli series converges. $\square$

Define the first threshold by

$$N(y) = \min\{n \geq 0 : p_n \geq y\} \quad (y > 0).$$

The sequence is nondecreasing: appending the new maximum $n+1$ extends the last ascending run and preserves both neighboring cross inequalities. This is injective, including from the empty permutation. The sequence is eventually strictly increasing by the ratio formula, and it tends to infinity.

For $L = \log y$ sufficiently large, there is a unique large real root $x = x(y)$ of $H(x) = L$, since

$$H'(x) = \psi(x+1) - \log \rho \sim \log x, \quad H''(x) = \psi'(x+1) > 0 \quad (x > -1).$$

We refer only to this eventually increasing branch.

Theorem 19.2 (Precise integer bracket). *Fix constants M, q, n_0 for which (19.1) holds for all integers $n \geq n_0$. Let x be the large root just defined. If*

$$x - 2 \geq n_0, \quad H'(x - 2) > 0, \quad \delta(x) = \frac{2Mq^{x-2}}{H'(x - 2)} < 1, \quad (19.5)$$

then

$$\lceil x - \delta(x) \rceil \leq N(y) \leq \lceil x + \delta(x) \rceil. \quad (19.6)$$

In particular $\delta(x) = O(q^x / \log x)$. If the two ceilings agree, they determine the threshold; otherwise exact counts or sharper certified information are needed.

Proof. Let $a = \lceil x - \delta \rceil - 1$ and $b = \lceil x + \delta \rceil$. Both belong to $[x - 2, x + 2]$, with $a < x - \delta$ and $b \geq x + \delta$. Since H' is increasing and positive there, the mean value theorem gives

$$H(x) - H(a) > H'(x - 2)\delta = 2Mq^{x-2}, \quad H(b) - H(x) \geq 2Mq^{x-2}.$$

Each logarithmic error at a, b is at most Mq^{x-2} because $q < 1$. Hence $p_a < y \leq p_b$. Monotonicity gives $a < N(y) \leq b$, which is the claim. The stated order of δ follows from $H'(x - 2) \sim \log x$. $\square$

This theorem is an exact implication once its constants and hypotheses are supplied. It is not yet a numerically executable asymptotic certificate: this report does not give effective certified M, q, n_0, ρ, C . The companion's bounded threshold search uses actual integer counts and has a different, finite scope.

19.1 Lambert initialization and every fixed inverse order

Let W be the positive real branch of Lambert's function, $W(s)e^{W(s)} = s$, and put

$$x_0 = \frac{L}{W(L/(e\rho))}, \quad u = \log(x_0/\rho), \quad b = \frac{1}{2} \log(2\pi x_0) + \log C. \quad (19.7)$$

Then $x_0 \log(x_0/(e\rho)) = L$ exactly. This supplies an initializer, not an exact threshold. Taylor expansion of H gives

$$x = x_0 - \frac{\frac{1}{2} \log(2\pi x_0) + \log C}{\log(x_0/\rho)} + O\left(\frac{1}{x_0 \log x_0}\right). \quad (19.8)$$

In particular $x - x_0 \rightarrow -1/2$. Taking a bare ceiling of x_0 is therefore not justified even by an absolute $o(1)$ approximation.

Here is an explicit recursive specification to all fixed orders. In formal power series in t , with u, b treated as parameters, define

$$\begin{aligned} \mathcal{F}(d, t; u, b) &= ud + b + \sum_{k \geq 2} \frac{(-1)^k d^k t^{k-1}}{k(k-1)} + \frac{1}{2} \log(1 + td) \\ &\quad + \sum_{j \geq 1} \frac{B_{2j} t^{2j-1}}{2j(2j-1)(1+td)^{2j-1}}. \end{aligned} \quad (19.9)$$

Every fixed coefficient uses only finitely many displayed terms. Set

$$\begin{aligned} d_0(u, b) &= -b/u, \\ d_m(u, b) &= -\frac{1}{u}[t^m]\mathcal{F}\left(\sum_{j=0}^{m-1} d_j(u, b)t^j, t; u, b\right) \quad (m \geq 1). \end{aligned} \quad (19.10)$$

These are rational functions of u, b ; for example,

$$\begin{aligned} d_1 &= -\frac{d_0^2/2 + d_0/2 + 1/12}{u}, \\ d_2 &= -\frac{(d_0 + 1/2)d_1 - d_0^3/6 - d_0^2/4 - d_0/12}{u}. \end{aligned} \quad (19.11)$$

Proposition 19.3 (All fixed orders of the continuous inverse). *For each fixed integer $m \geq 0$, with the parameters of (19.7),*

$$x = x_0 + \sum_{j=0}^m \frac{d_j(u, b)}{x_0^j} + O\left(\frac{x_0^{-m-1}}{\log x_0}\right) \quad (y \rightarrow \infty). \quad (19.12)$$

The result concerns the continuous root x ; no fixed truncation is an exact integer rounding rule for $N(y)$.

Proof. Here $b = u/2 + \frac{1}{2}\log(2\pi\rho) + \log C$, so d_0 is bounded. Induction in (19.10) gives $d_j = O(1/u)$ for every fixed $j \geq 1$: the new coefficient in the nonlinear terms is a finite polynomial in previously bounded d_j , and division by u remains. For bounded real d , finite Taylor expansion of $(x_0 + d)\log((x_0 + d)/(e\rho))$, of $\frac{1}{2}\log(2\pi(x_0 + d))$, and of a sufficiently long finite Stirling expansion shows that

$$H(x_0 + d) - L$$

has the fixed-order expansion (19.9) with $t = x_0^{-1}$. After truncation through degree m its remainder is $O(t^{m+1})$ uniformly for bounded d . The only unexpanded term with the large parameter u is the exact linear term ud . Substitute $d = \sum_{j=0}^m d_j t^j$; the defining recurrence cancels the coefficients through degree m , leaving residual $O(x_0^{-m-1})$.

To locate the root on the same bounded neighborhood, note that $H(x_0) - L = b + O(x_0^{-1})$ and, for example, $H(x_0 - 2) - L = b - 2u + O(x_0^{-1}) < 0$ for large x_0 , whereas $H(x_0) - L > 0$. Thus $x \in (x_0 - 2, x_0)$. The proposed approximation also lies in a fixed bounded neighborhood of x_0 . On the interval between the two, $H' \asymp \log x_0$. The mean value theorem converts the residual into (19.12). For $m = 0$ it gives (19.8). $\square$

The integer inverse has $N(y) = x + O(1)$ and the stronger ceiling bracket (19.6); an ordinary algebraic expansion cannot remove rounding at thresholds arbitrarily close to integers. Neither differentiating an unknown remainder nor silently rounding an approximation is used here.

[write] Corollary 19.1 and Theorem 19.2 are Part I's (11.2) and Proposition 11.1, with the same proof; (19.8) is Part I's (11.4), because this Part's u is Part I's u plus one. Proposition 19.3 goes beyond Part I. The front matter ("The inverses and the transseries volume") shows that the recursion (19.10) is an instance of the transseries volume's reversion about a core, and classifies the other inverse statements.

20 Finite checks and source position

The companion checks combinatorial and algebraic inputs by exact integer arithmetic. Such checks neither replace the infinite proof nor certify a compact-operator discretization. The main proof does not rely on numerical fitting or on a discretized eigenvalue.

The prior endpoint recurrence [3] can be stated as follows. Let $f_n(c, d)$ count admissible permutations whose last ascending run has minimum c and maximum d , and define the rectangular sum

$$F_n(a, b) = \sum_{1 \leq c \leq a} \sum_{b \leq d \leq n} f_n(c, d),$$

with out-of-range endpoints giving the natural empty sum. For $n = 1$, $f_1(1, 1) = 1$. For $n \geq 2$,

$$f_n(c, c) = F_{n-1}(c-1, c), \quad (20.1)$$

$$f_n(c, d) = \mathbf{1}\{c = 1, d = n\} + \sum_{\ell=0}^{d-c-1} \binom{d-c-1}{\ell} F_{n-\ell-2}(d-\ell-2, c) \quad (c < d), \quad (20.2)$$

where terms with $n - \ell - 2 < 1$ in the sum are zero. To justify the indices, delete the final run, whose ℓ interior values are chosen from the $d - c - 1$ values between c, d . After relabelling, the preceding run minimum must be at most $d - \ell - 2$ and its maximum at least c . These are precisely the two overlap constraints. The isolated all-increasing permutation supplies the first term. For a singleton final run the preceding minimum is at most $c - 1$ and maximum at least c after relabelling. Rectangle sums are formed by two-dimensional prefix/suffix accumulation, and $p_n = F_n(n, 1)$.

A structurally independent test uses every composition of n as a proposed maximal-run composition. It creates a poset on the actual positions by imposing internal ascents and both boundary cross inequalities. Exact subset dynamic programming counts its linear extensions (zero if constraints are cyclic). Summing over all compositions verifies directly the labelled-simplex normalization and uniqueness of the run decomposition. Another test enumerates all input permutations, applies the literal decreasing-run reversal, and compares its image as a set with the overlap predicate as a set.

The released finite-check packet records:

- exact endpoint-recurrence counts through $n = 70$;
- agreement with all 25 terms displayed by the inspected OEIS entry and all 45 tabulated terms preserved from the counting paper;
- literal image and overlap-set equality through $n = 9$;
- independent position-poset totals through $n = 10$;
- append-maximum injectivity and the cross-inequality edge cases, including the admissible decreasing-triple example 14325.

The code has explicit input bounds, deterministic JSON output, standard-library dependencies only, and validation active with Python's optimization switch. The exact threshold command searches only its bounded computed count prefix and reports an unreachable threshold rather than substituting an asymptotic value.

20.1 What the source gate establishes

The inspected OEIS entry A307030 [9] defines adjacent overlapping runs, gives a 25-term prefix, and links the primary papers. Its saved official-data version is number 63, dated 18 November 2021; it was retrieved afresh on 3 October 2026. The absence of a displayed asymptotic formula alone is not evidence of novelty.

The 2019 image paper [1] provides the characterization and factorial-order lower bounds. Claesson, Guðmundsson and Pantone [3, Section 3.3] give numerical predictions for the leading constants, further positive and complex poles, and non-D-finiteness. The chronology is a 2019 preprint and a 2021 online DOI; some final-volume metadata is dated 2023. We do not infer an unseen later revision. Asinowski, Banderier and Hackl [2, Section 2.4] explicitly identify exact asymptotic analysis as a challenge and repeat the numerical law. The relevant enumeration section of *Semidistrim Lattices* [5], including its A307030 table, does not provide this equivalent.

Ehrenborg, Kitaev and Perry [6] use uniform-coordinate normalization, compact integral operators, positivity, and spectral asymptotic expansions for consecutive-pattern avoidance. Those methods are established prior art. Their fixed-window transfer operator differs from the present mixed point/interval, variable-run resolvent. The sequence-specific input here consists of that representation, its positive simple crossing, and the strict complex comparison that isolates the dominant pole.

The bounded search additionally checked combinations of “pop-stacked”, “overlapping runs”, A307030, asymptotic, spectral, Fredholm, and integral terminology, as well as later dated references and the available report catalogue. It found no competing proof of the total-count equivalent. A claim concerning the separate fixed-number-of-runs denominator conjecture is a different problem. The conclusion is a new-proof claim relative to the sources checked, not a guarantee about every unpublished or inaccessible work.

20.2 Reproducibility and remaining limits

The archive includes this editable $\text{\LaTeX}$ source, the PDF, a bounded exact companion and tests, numerical count data, source provenance, and fixed-allowlist PDF/archive replay tools. It excludes full third-party papers. Two clean PDF replays and two archive replays are required for the release; deterministic byte equality is a same-toolchain guarantee, not a cross-platform one. The README records commands and the toolchain. Output paths are created exclusively; source trees and output parents must be trusted and nonadversarial. These checks are reproducibility safeguards rather than a sandbox for modified code or $\text{\TeX}$.

The following remain outside the theorem: certified decimal intervals for ρ, C ; effective values for R, M, n_0 ; an exact finite-input asymptotic threshold certificate; the conjectured additional poles; and non-D-finiteness. None is needed for the factorial-exponential equivalent or its fixed-order consequences.

[write] 6 October 2026. Two of these limits are removed by Part I: certified decimal intervals for ρ and C (10.1), and the additional poles with non-D-finiteness (Sections 5–6). Effective R, M, n_0 and a finite-input threshold certificate remain open (Section 33, items 1 and 2). The “separate fixed-number-of-runs denominator conjecture” mentioned above is Conjecture 2 of [3]; it is now proved, first by Patel [11] and then by Part III, Theorem 30.1.

Part III

Run distribution in pop stacked permutations: an explicit bivariate generating function and Gaussian laws

Bundle Report 166, archive `Pop_Stacked_Permutations_Run_Distribution_Source.zip`, delivered as `Report166.tex` (16 pages). Delivered title block “Run distribution in pop stacked permutations / An explicit bivariate generating function and Gaussian laws / Report 166 / 3 October 2026”. Printed as delivered, with [write] notes; Report 166’s Section k is Section $k + 20$ here, and its statements and equations move with their sections. The opening paragraph “Reading and scope” is the manuscript’s own.

Abstract of this Part (as delivered). Let $p_{n,k}$ count distinct permutations of $[n]$ in the image of a single pop-stack pass, with k maximal ascending runs. We derive an explicit bivariate exponential generating function $P(z, u) = \sum_{n,k} p_{n,k} u^k z^n / n!$ from a weighted mixed-state integral model. Endpoint differentiation, a sufficient Robin boundary problem, and an explicit cancellation give a single-valued meromorphic formula.

For a uniformly chosen distinct image permutation, the number of runs has mean $\mu n + \beta + O(\eta^n)$ and variance $vn + \gamma + O(\eta^n)$, with $0 < \eta < 1$, $\mu \approx 0.44214954918886$ and $v \approx 0.06008555058270$. We give exact analytic definitions of all four constants and rational certificates for both slopes. Complex-uniform singularity transfer proves a central limit theorem. A spectral aperiodicity argument and Fourier inversion prove a uniform lattice local limit with absolute error $O(n^{-1})$.

As a separate application, a filtered expansion proves the fixed-run denominator representation and exact numerator degree conjectured by Claesson, Guðmundsson and Pantone. Shivam Patel publicly posted a correct inclusion–exclusion proof on 20 August 2026; the present application is an alternate proof. The site labels that post an unverified claim, which is distinct from our independent mathematical assessment. No first-resolution claim is made.

Reading and scope. The probabilistic object is uniform on distinct outputs, not the image of a uniformly random input. The derivation and complex estimates are standalone. The companion supplies bounded exact checks, not substitutes for proofs. The bivariate formula and Gaussian laws go beyond the sources checked; the search does not certify worldwide priority. Reports 161 and 164 remain unchanged. No global bivariate pole classification, minimal fixed-run denominator, refined-count complexity improvement, Edgeworth expansion, or effective numerical asymptotic error onset is claimed.

Reading conventions of Part III [write]

$p_{n,k}, p_n(u), P(z, u), K_n$	A309993, row polynomials, the bivariate EGF ($P(z, 1)$ is Part I’s P), the run count (a random variable, not a kernel)
$Q, B, h_0, H, J_0, D(z, u)$	(21.1)–(21.2) and $BH - QJ_0$; at $u = 1$ Part I’s Q, B, e^z, H, J_0, D
K, ν, Δ	overlap kernel, measure (Part I’s μ), interval states
$\mu, v, \beta, \gamma, \eta$	mean and variance slopes, constant terms, error ratio (μ is not a measure here)
$\rho(u), C(u), \Phi, H_c, G_c$	moving pole, its amplitude, local phase, cumulant functions
F_k, N_k, D_k, q, δ	fixed-run OGF, numerator, denominator; $q = e^z$ and δ as formal objects in Section 30

21 Counting model and main results

A pop-stack pass reverses each maximal decreasing run and concatenates the resulting increasing blocks. Its distinct output permutations are called *pop-stacked*. Let $p_{n,k}$ count those of size n with k maximal ascending runs. Put

$$p_n(u) = \sum_{k \geq 0} p_{n,k} u^k, \quad P(z, u) = \sum_{n \geq 0} p_n(u) \frac{z^n}{n!}, \quad p_0(u) = 1.$$

Thus the empty permutation has zero runs. For $n \geq 1$, let K_n be the run count of a permutation chosen uniformly from the $p_n(1)$ distinct outputs. Then $\mathbb{E}u^{K_n} = p_n(u)/p_n(1)$. Different input permutations can produce the same output; weighting by their preimage sizes gives a different distribution and is not considered here.

Define

$$\begin{aligned} Q &= 4u + 4u^2(e^z - 1) - 1, & E &= e^{z/2}, \\ B &= 2uE + 2u - 1, & h_0 &= 1 + u(e^z - 1), \end{aligned} \quad (21.1)$$

$$H(z, u) = \sum_{j \geq 0} \frac{(-1)^j z^{2j} Q^j}{16^j (2j)!}, \quad J_0(z, u) = \frac{z}{4} \sum_{j \geq 0} \frac{(-1)^j z^{2j} Q^j}{16^j (2j+1)!}. \quad (21.2)$$

The last two series converge uniformly on compact subsets of $\mathbb{C}^2$.

Theorem 21.1 (Bivariate exponential generating function). *As a germ at $z = 0$, and by single-valued meromorphic continuation,*

$$\boxed{P(z, u) = \frac{Q(H + BJ_0)}{h_0(BH - QJ_0)}}. \quad (21.3)$$

Removable quotient presentations are filled in wherever the counting germ is analytic. Locally near $(z, u) = (0, 1)$ choose $r = \sqrt{Q}$ with $r(0, 1) = \sqrt{3}$ and put $\theta = zr/4$. Then equivalently

$$P(z, u) = \frac{r}{h_0} \frac{r \cos \theta + B \sin \theta}{B \cos \theta - r \sin \theta}. \quad (21.4)$$

For instance, $P(0, u) = 1$ and $[u]P = e^z - 1$. At exceptional parameters such as $u = 1/4$, the displayed numerator and denominator can both vanish at $z = 0$; the counting series still supplies the analytic value. We do not assert that every zero of the written denominator is a pole.

[write] At $u = 1$ the definitions give $Q = 4e^z - 1$, $B = 2e^{z/2} + 1$ and $h_0 = e^z$, so (21.3) is Part I's (1.5), and (21.4) is Part I's (1.6): a second derivation of Part I's formula. At $u = 1$ Part I answers the question left open in the last sentence: the genuine poles of $P(z, 1)$ are exactly the zeros of $D(z, 1)$ with $Q \neq 0$ (Part I, (1.7)).

Let ρ be the unique positive solution of

$$\frac{\rho \sqrt{4e^\rho - 1}}{4} + \arctan \frac{\sqrt{4e^\rho - 1}}{2e^{\rho/2} + 1} = \frac{\pi}{2}, \quad \rho = 1.113439041736727 \dots \quad (21.5)$$

The arctangent here is the positive real branch. Section 25 proves that ρ is the unique dominant univariate pole. Write $x = \rho$, $T = e^\rho$. Define

$$\mu = \frac{2Tx - x - 1}{Tx(x + 2)}, \quad (21.6)$$

$$v = \frac{4T^2x^3 - 3Tx^4 - 10Tx^3 - 2Tx^2 + 4Tx + x^4 + 4x^3 + 5x^2 + 4x + 2}{T^2x^2(x + 2)^3}. \quad (21.7)$$

Theorem 21.2 (Moments and Gaussian laws). *There are real constants β, γ and $0 < \eta < 1$ such that*

$$\mathbb{E}K_n = \mu n + \beta + O(\eta^n), \quad \text{Var } K_n = v n + \gamma + O(\eta^n). \quad (21.8)$$

The constants β, γ are defined analytically in (27.4). The variance slope v is strictly positive, and

$$\frac{K_n - \mu n}{\sqrt{vn}} \implies \mathcal{N}(0, 1). \quad (21.9)$$

Moreover, uniformly over every integer k ,

$$\sup_{k \in \mathbb{Z}} \left| \mathbb{P}(K_n = k) - \frac{\exp(-(k - \mu n)^2 / (2vn))}{\sqrt{2\pi vn}} \right| = O(n^{-1}). \quad (21.10)$$

The error in (21.10) is absolute, not a uniform relative error in the tails. Its constants and onset are not numerically certified. Using the exact mean and variance also yields (21.9); replacing μn by $\mu n + \beta$ in (21.10) preserves its stated accuracy.

22 Run weights and factorial normalization

The established overlapping-run characterization [1, 2] says that consecutive maximal ascending runs with endpoints (a, b) and (c, d) satisfy

$$a < d, \quad c < b. \quad (22.1)$$

The second inequality is the descent between runs; the first is the extra image condition. A singleton has equal endpoints, so it can occur strictly inside the range of a neighboring nonsingleton run, but two singletons cannot be adjacent.

Use the disjoint union

$$I = (0, 1), \quad \Delta = \{(a, b) : 0 < a < b < 1\}, \quad X = I \sqcup \Delta, \quad \nu = dx \sqcup da db.$$

A state in I has $\text{lo}(x) = \text{hi}(x) = x$; an interval state has $\text{lo} = a, \text{hi} = b$. Define the symmetric overlap kernel and weights

$$K(s, t) = \mathbf{1}\{\text{lo}(s) < \text{hi}(t), \text{lo}(t) < \text{hi}(s)\}, \quad w_z(x) = z, \quad w_z(a, b) = z^2 e^{z(b-a)}, \quad (22.2)$$

$$(A_z f)(s) = \int_X K(s, t) w_z(t) f(t) d\nu(t). \quad (22.3)$$

On $L^2(X, \nu)$, A_z is compact and entire in operator norm: its bounded kernel is square integrable, and the weights and their series are uniformly bounded on compact z -sets. Also $\|A_z\| = O(|z|)$ near zero.

For a run of length $\ell \geq 2$ with endpoints $a < b$, the ordered interior coordinates occupy volume $(b - a)^{\ell-2} / (\ell - 2)!$. Summing z^ℓ times this volume gives $w_z(a, b)$. Every rank order of n distinct coordinates has volume $1/n!$. Thus the product of weights and kernels counts precisely the labeled output permutations with EGF normalization. No run-order factorial is needed. The check

$$\int_X w_z d\nu = z + z^2 \int_0^1 (1 - t) e^{zt} dt = e^z - 1$$

recovers the unique one-run permutation in each positive size.

Multiplying every run weight by u marks the number of runs exactly once. Summing the chain of run states gives

$$h = (I - uA_z)^{-1}\mathbf{1}, \quad P(z, u) = 1 + u \int_X w_z(s)h(s) d\nu(s). \quad (22.4)$$

The pairing in this formula is bilinear. For bounded u and sufficiently small z , the Neumann series converges uniformly, proving joint analyticity and the counting identity. Each $p_n(u)$ has degree at most n . Integral-operator enumeration has classical precedents, for example [6]; the particular state space and calculations are given here.

[write] This is the model of Part I, Section 2, and Part II, Section 15, with every run weight multiplied by u .

23 Endpoint equations and a sufficient boundary problem

For small z set

$$\begin{aligned} U(x) &= u \int_{\text{hi}(s) \leq x} w_z(s)h(s) d\nu(s), & V(x) &= u \int_{\text{lo}(s) < x} w_z(s)h(s) d\nu(s), \\ T_0 = P - 1 &= U(1) = V(1), & U(0) &= V(0) = 0. \end{aligned}$$

Boundary equality sets have measure zero. Complementing the two nonoverlap regions yields

$$h(x) = 1 + V(x) - U(x), \quad h(a, b) = 1 + V(b) - U(a). \quad (23.1)$$

These identities provide continuous representatives and then the smoothness needed for differentiation. Put

$$I_0(x) = \int_0^x e^{z(x-a)} U(a) da, \quad J_1(x) = \int_x^1 e^{z(b-x)} V(b) db.$$

Differentiating the endpoint cumulatives gives

$$U' = uz e^{zx} (1 + V) - uzU - uz^2 I_0, \quad I_0' = U + zI_0, \quad (23.2)$$

$$V' = uzV + uz e^{z(1-x)} (1 - U) + uz^2 J_1, \quad J_1' = -V - zJ_1. \quad (23.3)$$

For example, the interval part of U' is $uz^2 \int_0^x e^{z(x-a)} (1 + V(x) - U(a)) da$; combining its constant term with the singleton contribution gives (23.2). Differentiating once more cancels the integrals. With $p = U'$, $q = V'$,

$$\begin{aligned} p' &= (1 - u)zp + uz e^{zx} q, & p(0) &= uz, \\ q' &= (u - 1)zq - uz e^{z(1-x)} p, & q(1) &= uz. \end{aligned} \quad (23.4)$$

The last boundary condition uses $U(1) = V(1)$ and is essential. Eliminating q , locally near $u = 1$, gives the Robin problem

$$\begin{aligned} p'' - zp' + z^2 \kappa p &= 0, & \kappa &= u + u^2(e^z - 1), \\ p(0) &= uz, & p'(1) - (1 - u)zp(1) &= u^2 z^2 e^z. \end{aligned} \quad (23.5)$$

At $z = 0$, the homogeneous two-point first-order system has only the zero solution. Analytic dependence of its fundamental matrix gives local existence and uniqueness, uniformly near $(z, u) = (0, 1)$.

Lemma 23.1 (Sufficiency). *The local solution of (23.4) reconstructs the counting resolvent (22.4); differentiation has lost no condition.*

Proof. Reflection sends $(p(x), q(x))$ to $(q(1-x), p(1-x))$ with the same system and boundary data. Uniqueness therefore gives $p(x) = q(1-x)$. Integrating p, q from zero yields U, V with $U(1) = V(1)$. Define I_0, J_1 as above and the residuals

$$\begin{aligned} R_1 &= p - uze^{zx}(1+V) + uzU + uz^2I_0, \\ R_2 &= q - uzV - uze^{z(1-x)}(1-U) - uz^2J_1. \end{aligned}$$

Direct use of (23.4) gives $R'_1 = zR_1$, $R'_2 = -zR_2$, with $R_1(0) = 0$ and $R_2(1) = uz(U(1) - V(1)) = 0$. Thus both vanish. The endpoint cumulatives of the h reconstructed from (23.1) have derivatives U', V' and zero initial values; they equal U, V . The overlap complement then gives $h = 1 + uA_z h$. Local Neumann uniqueness identifies this h with (22.4). $\square$

Apparent divisions by z in the scalar elimination are used for nonzero z first and extended analytically to zero.

24 Explicit solution and half angle cancellation

Write $s = 2u - 1$, $b = 2ue^{z/2}$, $r = \sqrt{Q}$, and $\theta = zr/4$. Then $Q = b^2 - s^2$ and $B = b + s$. Solving (23.5) gives

$$\begin{aligned} p(x) &= e^{zx/2} \{uz \cos(zrx/2) + A \sin(zrx/2)\}, \\ a &:= \frac{A}{uz} = \frac{b - s \cos(2\theta) + r \sin(2\theta)}{r \cos(2\theta) + s \sin(2\theta)}. \end{aligned} \tag{24.1}$$

For $c = \cos \theta$, $d = \sin \theta$ define

$$D_+ = (b+s)c - rd, \quad D_- = (b-s)c + rd, \quad L_0 = rc + (b+s)d.$$

Direct multiplication, using $r^2 = b^2 - s^2$, proves

$$\begin{aligned} D_+ D_- &= r \{r \cos(2\theta) + s \sin(2\theta)\}, \\ D_- L_0 &= r \{b - s \cos(2\theta) + r \sin(2\theta)\}. \end{aligned} \tag{24.2}$$

At $z = 0$, $D_- = 1$. The common factor cancels in the germ and $a = L_0/D_+$. Zeros of the canceled factor alone are not scalar poles.

Reflection and the first equation of (23.4) at zero give $p(1) = q(0) = z(s + ar)/2$. Integrating (23.5), with both endpoint derivatives retained, gives

$$z^2 \kappa T_0 = 2uzp(1) - u^2 z^2 (e^z + 1), \quad P = 1 + T_0 = \frac{ura}{\kappa} = \frac{ra}{h_0}.$$

This proves (21.4) including its prefactor. The local identities $H = \cos \theta$, $J_0 = \sin \theta / r$ prove (21.3). The entire definitions in (21.2) give a single-valued meromorphic continuation. The proof does not choose a global square root or classify the global bivariate polar set.

25 Isolation of the dominant univariate pole

We give the spectral isolation needed for a complex parameter transfer, not only a real-root calculation. Let $\mathcal{K}$ have kernel K . For real $t > 0$, A_t is similar to the compact self-adjoint operator

$$S_t = M_{\sqrt{w_t}} \mathcal{K} M_{\sqrt{w_t}}.$$

The multiplier and its inverse are bounded. The operator preserves positivity but need not be positive semidefinite. Every pair of states s, s' is overlapped by all intervals satisfying

$$0 < a < \min(\text{lo}(s), \text{lo}(s')), \quad \max(\text{hi}(s), \text{hi}(s')) < b < 1.$$

This is a positive-area set; hence S_t^2 has a strictly positive kernel almost everywhere.

Lemma 25.1. *The spectral radius $\lambda(t)$ of A_t is a simple positive eigenvalue with positive left and right eigenfunctions. All its other eigenvalues have strictly smaller modulus.*

Proof. The compact self-adjoint spectral theorem gives a top eigenvector of S_t^2 . Its absolute value cannot lower its Rayleigh quotient and is therefore also a top eigenvector. Strict kernel positivity makes that vector positive almost everywhere. A sign-changing top vector would strictly increase its quotient upon taking absolute values, which is impossible. Two independent real top vectors would have a nonzero linear combination orthogonal to the positive one and thus changing sign. The top eigenspace is consequently one-dimensional. Positivity of S_t selects the eigenvalue $+\|S_t\|$ on it. Any other eigenvalue of the same modulus would give another top vector of S_t^2 . Similarity transfers the conclusion to A_t , and compactness gives the gap. $\square$

Continuity and strict increase of the weights give continuity and strict increase of $\lambda(t)$, by testing the Rayleigh quotient on its positive Perron vector. It tends to zero as $t \downarrow 0$. On $F = \{(a, b) : a < 1/4, b > 3/4\}$, every pair overlaps and $w_t \geq t^2 e^{t/2}$; the indicator Rayleigh quotient gives $\lambda(t) \geq t^2 e^{t/2}/16 \rightarrow \infty$. There is a unique $t_* > 0$ with $\lambda(t_*) = 1$.

A Riesz projection at this isolated simple eigenvalue gives local analytic dependence, as in standard perturbation theory [8]. For the real unit Perron eigenvector f_t of S_t ,

$$\lambda'(t) = \lambda(t) \int_X \frac{w'_t}{w_t} f_t^2 d\nu > 0. \quad (25.1)$$

Indeed differentiating S_t gives $S'_t = \frac{1}{2}(M_{w'_t/w_t} S_t + S_t M_{w'_t/w_t})$, and the simple-eigenvalue derivative formula proves (25.1). The left and right A_{t_*} eigenvectors are $\ell_* = \sqrt{w_{t_*}} f_{t_*}$ and $h_* = f_{t_*}/\sqrt{w_{t_*}}$, paired to one. The scalar resolvent's singular numerator is

$$\left(\int_X \sqrt{w_{t_*}} f_{t_*} d\nu \right)^2 > 0.$$

Thus t_* is a genuine simple pole of $P(z, 1)$. For $|z| < t_*$, kernel comparison and the spectral-radius formula give $\text{spr}(A_z) \leq \lambda(|z|) < 1$, so there are no poles inside.

At $u = 1$, (21.4) becomes $P(z, 1) = e^{-z} r \tan \phi(z)$, where

$$\phi(z) = \frac{zr}{4} + \arctan \frac{r}{2e^{z/2} + 1}, \quad \phi'(z) = \frac{e^z(z+2)}{2r} > 0 \quad (z \geq 0).$$

Since $\phi(0) = \pi/6$ and $\phi(z) \rightarrow \infty$, its first positive pole is the ρ in (21.5). The preceding paragraph identifies $t_* = \rho$.

For $|z| = \rho$ with $z \neq \rho$, let $B_z^{\text{abs}} = \mathcal{K}M_{|w_z|}$. The same positive-kernel argument gives a positive Perron vector h_z with eigenvalue $b_z = \text{spr}(B_z^{\text{abs}})$. On the interval component,

$$|w_z(a, b)| = \rho^2 e^{\text{Re}(z)(b-a)} < w_\rho(a, b).$$

The singleton absolute weights are equal. Pairing with $\ell_* > 0$ gives

$$(1 - b_z) \int_X \ell_* h_z d\nu = \int_\Delta (w_\rho - |w_z|) h_z (\mathcal{K}\ell_*) d\nu > 0,$$

since $\mathcal{K}\ell_* = \ell_*/w_\rho > 0$ almost everywhere. Therefore $\text{spr}(A_z) \leq b_z < 1$ at every other point of the circle. The scalar resolvent is analytic there. Meromorphic discreteness of the explicit quotient now supplies a radius larger than ρ containing no other scalar pole.

[write] The same isolation as Part I, Section 9, and Part II, Sections 16–18; (21.5) is the case $k = 0$ of Part I's (5.4) (Part I writes the phase with $\frac{1}{2} \arccos(1/(2e^{z/2}))$, equal to the arctangent here for $z \geq 0$).

26 Complex uniform transfer with a moving pole

The rational certificate in Section 28 implies $\rho < \log 4$. Choose R with $\rho < R < \log 4$ sufficiently close to ρ . At $u = 1$, $Q = 4e^z - 1$ and $h_0 = e^z$ do not vanish on $|z| \leq R$. Put

$$D(z, u) = BH - QJ_0.$$

No zero of $D(z, 1)$ in this disk cancels against the numerator: on a local square-root branch the two linear combinations in (21.4) have determinant

$$B^2 + r^2 = 4e^{z/2}(2e^{z/2} + 1) \neq 0.$$

Here $B = 0$ would imply $Q = 0$. Simultaneous vanishing would force sine and cosine to vanish together. The spectral result therefore implies that $D(z, 1)$ has exactly the one simple zero ρ in the disk, and none on its boundary.

For complex u sufficiently near one, compactness keeps Q and h_0 nonzero throughout the closed disk. Rouché's theorem preserves exactly one zero of D , counting multiplicity. The implicit function theorem gives its holomorphic position $\rho(u)$, with $\rho(1) = \rho$. Its numerator and z derivative remain nonzero. Analytic division at the moving simple zero consequently gives

$$P(z, u) = \frac{C(u)}{1 - z/\rho(u)} + G(z, u), \quad (26.1)$$

where C is holomorphic and nonzero, and G is jointly holomorphic on a neighborhood of the closed z -disk times a smaller u -disk. Cauchy's estimate on the fixed circle proves

$$\frac{p_n(u)}{n!} = C(u)\rho(u)^{-n} + O(R^{-n}), \quad \text{uniformly for complex } u \text{ near } 1. \quad (26.2)$$

The complex neighborhood and uniform error are important: pointwise real-parameter estimates alone would not justify moment differentiation. This is the usual singularity-perturbation mechanism

underlying quasi-powers [7, Chapter IX]; the needed uniform argument is supplied here rather than assumed from a real-variable version.

For explicit local formulas choose the analytic phase near $(\rho, 1)$

$$\Phi(z, u) = \frac{zr}{4} + \arctan \frac{r}{B}.$$

Choose its branch continuously from (21.5). Then $P = (r/h_0) \tan \Phi$, $\Phi(\rho(u), u) = \pi/2$, and

$$\begin{aligned}\Phi_z &= \frac{u^2 e^z (z+2) - (u-1)(2u-1)}{2r}, \\ \Phi_u &= \frac{z\{2u(e^z - 1) + 1\} - 1/u}{2r}.\end{aligned}\tag{26.3}$$

In particular $\Phi_z(\rho, 1) = e^\rho(\rho+2)/(2r) > 0$. Writing $N(z, u) = u^2 e^z (z+2) - (u-1)(2u-1)$, the residue gives

$$C(u) = \frac{2Q(\rho(u), u)}{\rho(u)h_0(\rho(u), u)N(\rho(u), u)}.\tag{26.4}$$

These formulas use a local phase only; no inverse trigonometric branch has been selected globally.

27 Exact moments and the variance slope

Define, for complex t near zero,

$$H_c(t) = \log \frac{\rho}{\rho(e^t)}, \quad G_c(t) = \log \frac{C(e^t)}{C(1)},\tag{27.1}$$

with logs zero at $t = 0$. Shrink the disk so that $|\rho(e^t)|/R$ is uniformly less than one and $C(e^t)$ stays nonzero. Dividing (26.2) by its value at $u = 1$ yields

$$\mathbb{E}e^{tK_n} = \exp\{nH_c(t) + G_c(t)\}(1 + O(\eta^n))\tag{27.2}$$

uniformly there, for some $\eta < 1$. The relative remainder is holomorphic. For large n , its analytic logarithm is uniformly $O(\eta^n)$ as well. Cauchy estimates on a smaller disk permit differentiation at zero. Thus (21.8) holds with

$$\begin{aligned}\mu &= H'_c(0) = -\frac{\rho'(1)}{\rho}, \\ v &= H''_c(0) = -\frac{\rho''(1) + \rho'(1)}{\rho} + \left(\frac{\rho'(1)}{\rho}\right)^2,\end{aligned}\tag{27.3}$$

$$\beta = G'_c(0), \quad \gamma = G''_c(0).\tag{27.4}$$

Equations (21.5), (26.4), (27.1) and (27.4) are exact analytic definitions of the constant terms. For orientation only, numerical differentiation gives

$$\beta \approx 0.3035533256498590630, \quad \gamma \approx 0.0084946050872070072.$$

These two decimal evaluations are not interval-certified. The slope certificates below do not certify the displayed β, γ digits.

[write] **6 October 2026.** An independent numerical check by the write (50 digits, not interval-certified): $\rho(u)$ from $D(z, u) = 0$ and $C(u)$ from (26.4), differentiated numerically in t with $u = e^t$, give μ and v equal to (21.6)–(21.7) to 25 digits, $\beta = 0.3035533256498590630377\dots$ and $\gamma = 0.008494605087207007156\dots$, agreeing with all displayed digits. The exact rows $p_n(u)$, $n \leq 40$, of `data/166-runs-data-full_verification.json` give $\mathbb{E}K_n - \mu n = 0.3035533271$, 0.30355332565 and 0.303553325649858 and $\text{Var } K_n - vn = 0.0084946100$, 0.00849460508846 and 0.00849460508720818 at $n = 20, 30, 40$, approaching β and γ as (21.8) predicts.

Here is the full derivative calculation of the slopes. Put $x = \rho$, $T = e^\rho$, $a_0 = 2T - 1$, $Q_0 = 4T - 1$, $M = xa_0 - 1$ and $N_0 = T(x + 2)$. All phase derivatives in the following display are at $(\rho, 1)$:

$$\begin{aligned} 2r\Phi_z &= N_0, \\ 2r\Phi_{zz} &= T(x + 3) - \frac{2T^2(x + 2)}{Q_0}, \\ 2r\Phi_{zu} &= 2T(x + 2) - 1 - \frac{2N_0a_0}{Q_0}, \\ 2r\Phi_{uu} &= 2x(T - 1) + 1 - \frac{2Ma_0}{Q_0}. \end{aligned} \tag{27.5}$$

Implicit differentiation of $\Phi(\rho(u), u) = \pi/2$ gives

$$\rho'(1) = -\frac{M}{N_0}, \quad \rho''(1) = -\frac{\Phi_{uu} + 2\Phi_{zu}\rho'(1) + \Phi_{zz}\rho'(1)^2}{\Phi_z}.$$

Substitution into (27.3) simplifies to (21.6)–(21.7). In particular positivity of v is not assumed on the basis of a fitted distribution or an informal quasi-powers claim; it is certified next.

28 Rational certificates for the root and slopes

The following terminating decimals denote exact rational endpoints:

$$\begin{aligned} L &= 1.11343904173672704376166152691808324014139016583344946615, \\ U &= 1.11343904173672704376166152691808324014139016583344946616, \\ \mu_L &= 0.442149549188859664032644356798334757622397538813, \\ \mu_U &= 0.442149549188859664032644356798334757622397538814, \\ v_L &= 0.060085550582698280075387627576274309378657258462, \\ v_U &= 0.060085550582698280075387627576274309378657258463. \end{aligned}$$

The exact certificate proves

$$L < \rho < U, \quad \mu_L < \mu < \mu_U, \quad 0 < v_L < v < v_U. \tag{28.1}$$

For $0 \leq x < 202$, the positive Taylor polynomial satisfies

$$T_{200}(x) := \sum_{j=0}^{200} \frac{x^j}{j!} \leq e^x \leq T_{200}(x) + \frac{x^{201}/201!}{1 - x/202}. \tag{28.2}$$

Indeed all successive ratios in the tail after its first term are at most $x/202$. Exponential monotonicity extends this to interval inputs. For rational $x \geq 0$, set $S = 10^{120}$ and compute $k = \lfloor \sqrt{\lceil xS^2 \rceil} \rfloor$ using integer square root. Checked integer squares give

$$(k/S)^2 \leq x < ((k + 1)/S)^2,$$

a rational enclosure for $\sqrt{x}$. On $0 \leq x \leq 1$, truncate the alternating sine and cosine series at index 60. The last included term is positive, giving an upper bound; subtracting the next term's magnitude gives a lower bound. Monotonicity of sine and cosine on this interval propagates interval inputs.

For $z = L, U$, these bounds enclose

$$D(z, 1) = (2e^{z/2} + 1) \cos(z\sqrt{4e^z - 1}/4) - \sqrt{4e^z - 1} \sin(z\sqrt{4e^z - 1}/4).$$

The program checks that the trigonometric arguments lie in $(0, 1)$, and that $D(L, 1) > 0 > D(U, 1)$. In this bracket the phase lies below $1 + \pi/4 < \pi$, so only its first positive pole can be crossed. Strict phase monotonicity isolates ρ . Also $U < 6/5 < \log 4$: (28.2) at $6/5$ gives $e^{6/5} < 4$.

Exact rational interval substitution in (21.6) and (21.7) then proves (28.1); in particular the entire numerator of (21.7) has a strictly positive lower bound. The output records outward decimal bounds for both root signs and propagated intervals. No floating-point fit or trusted transcendental decimal enters these checks. The suggested decimal endpoints become certificates only when the exact inequalities pass. They do not supply a numerical complex gap, an asymptotic error constant, or an onset index.

[write] The endpoints L, U are Part I's (10.1), digit for digit; the intervals for μ and v and the bridge $U < 6/5 < \log 4$ are new here. The write reran `run_checks.py` all on a fresh extraction; its output reproduced `data/166-runs-data-full_verification.json`.

29 Central and uniform lattice local limits

For fixed real y , substitute $t = iy/\sqrt{vn}$ in (27.2). Taylor expansion gives

$$n\{H_c(t) - \mu t\} = -y^2/2 + O_y(n^{-1/2}), \quad G_c(t) = O_y(n^{-1/2}).$$

The centered characteristic functions therefore converge to $e^{-y^2/2}$. The characteristic-function continuity theorem proves (21.9). The moment estimates show that exact mean and variance standardization gives the same limit.

The local limit also needs uniform control away from zero frequency. For a fixed $0 < \delta < \pi$, put $U_\delta = \{e^{i\theta} : \delta \leq |\theta| \leq \pi\}$. For $u \in U_\delta$ and $|z| < \rho$, the operator $I - uA_z$ is invertible because $\text{spr}(uA_z) < 1$. The strict comparison in Section 25 proves the same for $|z| = \rho$, $z \neq \rho$. At $z = \rho$, the only eigenvalue of A_ρ on the unit circle is its simple eigenvalue 1, whereas $1/u \neq 1$. Hence $I - uA_\rho$ is invertible as well.

Invertibility is open in operator norm and the inverse is continuous. Compactness of $\{|z| \leq \rho\} \times U_\delta$ therefore supplies $R_\delta > \rho$ such that $I - uA_z$ is invertible for every $|z| \leq R_\delta$, $u \in U_\delta$. Explicitly, otherwise a sequence of noninvertible pairs approaching that compact set would have a subsequence converging to an invertible point, contradicting openness. After decreasing R_δ slightly if needed, the boundary resolvents are uniformly bounded. Cauchy estimates in (22.4) and the positive leading coefficient asymptotic from (26.2) give

$$\sup_{\delta \leq |\theta| \leq \pi} |\mathbb{E} e^{i\theta K_n}| = O((\rho/R_\delta)^n). \quad (29.1)$$

This proves the needed lattice aperiodicity from the operator rather than guessing it from finite coefficient rows.

Near zero set $h(\theta) = H_c(i\theta) - i\mu\theta$. Since $v > 0$,

$$h(\theta) = -v\theta^2/2 + O(|\theta|^3), \quad \operatorname{Re} h(\theta) \leq -c\theta^2$$

for sufficiently small fixed $\delta > 0$. Also $e^{G_c(i\theta)} = 1 + O(|\theta|)$. The identity

$$e^A - e^B = (A - B) \int_0^1 e^{tA + (1-t)B} dt$$

with $A = nh(\theta)$, $B = -vn\theta^2/2$ gives, uniformly for $|\theta| \leq \delta$,

$$\left| e^{-i\mu n\theta} \mathbb{E} e^{i\theta K_n} - e^{-vn\theta^2/2} \right| \leq C(|\theta| + n|\theta|^3) e^{-c'n\theta^2} + C\eta^n. \quad (29.2)$$

All constants here are fixed. Fourier inversion on $[-\pi, \pi]$, together with the whole-line Fourier integral of the Gaussian, now proves (21.10). Indeed,

$$\int_{-\delta}^{\delta} |\theta| e^{-c'n\theta^2} d\theta = O(n^{-1}), \quad n \int_{-\delta}^{\delta} |\theta|^3 e^{-c'n\theta^2} d\theta = O(n^{-1}).$$

The omitted Gaussian tail and (29.1) are exponentially small. The remaining oscillatory factor has modulus one for every integer k , so the absolute error is uniform over all $\mathbb{Z}$. This is a direct proof of the claimed rate, with no appeal to a local limit theorem whose hypotheses have not been checked.

30 An alternate fixed run denominator proof

For fixed $k \geq 1$, let $F_k(x) = \sum_{n \geq 0} p_{n,k} x^n$ be the *ordinary* generating function. Fixed-run rationality was proved in [2, Theorem 6]. Claesson, Guðmundsson and Pantone [3, Conjecture 2] proposed the representation below.

Prior public proof. Shivam Patel posted an inclusion–exclusion proof on MathDB on 20 August 2026 at 14:02:52 UTC, edited at 14:04:08 UTC [11]. The public page labels it an unverified claimed solution. Reading the full argument, we find it valid for the denominator representation, exact degree, and leading coefficient stated here. This mathematical assessment is not a claim of peer review, formal verification, or community acceptance. The derivation below is a different proof and an application of the bivariate EGF, not a first resolution of that conjecture.

[write] **6 October 2026.** At intake (earlier on 6 October 2026) the MathDB page returned HTTP 403; at the write it was reachable, and the write read the posted solution in full. Its metadata give the posting and editing times stated above, the solution is signed Shivam Patel, and the site’s “Progress summary” marks the problem “Claimed solved” and says that the posted proof “has not been independently verified”. The write checked the argument step by step (summary in the front matter, “The predictions of Claesson, Guðmundsson and Pantone”): it is correct and complete for the denominator, the exact degree and the leading coefficient, as this Part says. It also proves $N_k(1) = 2(-1)^{k(k+1)/2-1} \prod_{j=1}^{k-1} j!$ for $k \geq 2$, which the write checked against $N_2, \dots, N_5$. Priority: Patel’s proof, posted 20 August 2026, precedes this Part (3 October 2026); this Part’s proof is a different one.

Theorem 30.1 (Fixed-run representation). *For every $k \geq 1$,*

$$F_k(x) = \frac{N_k(x)}{D_k(x)}, \quad D_k(x) = \prod_{j=1}^k (1 - jx)^{k-j+1}, \quad N_k \in \mathbb{Z}[x], \quad (30.1)$$

with $\deg N_k = k(k+1)/2$ and leading coefficient

$$(-1)^{k(k+1)/2+1} \prod_{j=1}^k j!.$$

No reduced or minimal denominator assertion is included.

Proof. Temporarily put $q = e^z$, $s = 2u - 1$, $L_1 = 1 + u(q - 1)$ and

$$\delta = \sqrt{(1 - 2u)^2 - 4u^2q}, \quad \delta|_{u=0} = 1.$$

Using $r = i\delta$ in the full-angle form (24.1) gives

$$P = \frac{\delta}{L_1} \frac{2ue^{z/2} - s \cosh(z\delta/2) - \delta \sinh(z\delta/2)}{\delta \cosh(z\delta/2) + s \sinh(z\delta/2)}.$$

Multiply numerator and denominator by $2e^{z\delta/2}$ and divide by $\delta - s$ to obtain

$$\begin{aligned} P &= \frac{\delta}{L_1} \frac{1 - A + T_1}{1 + A}, \\ A &= \frac{\delta + s}{\delta - s} qe^{z(\delta-1)}, \quad T_1 = \frac{4u}{\delta - s} qe^{z(\delta-1)/2}. \end{aligned} \quad (30.2)$$

This identity applies to the $u = 0$ counting germ. To justify moving from the derivation near $u = 1$, the Neumann model is jointly analytic for small z and bounded u . The branch-free quotient is meromorphically identical there by the identity theorem, so it continues to $u = 0$. Equivalently the full-angle expression is invariant under $r \mapsto -r$. At $u = 0$, $\delta - s = 2$ and $L_1 = 1$, so (30.2) is analytic as a u -series; apparent intermediate square-root presentations cause no change in the counting branch.

Now regard q, z as independent formal variables. Let $\mathcal{R}$ be the ring of formal u -series whose u^k coefficient is in $\mathbb{Q}[q, z]$ and has total degree at most k . A member with nonzero rational constant coefficient has its inverse in $\mathcal{R}$, by the coefficient recurrence. If $F \in \mathcal{R}$ has zero u -constant term, then $\exp F \in \mathcal{R}$ as well, because each coefficient is a finite sum of products and total degrees add.

The expansion

$$\delta = (1 - 2u) \sum_{m \geq 0} \binom{1/2}{m} (-4u^2q)^m (1 - 2u)^{-2m}$$

shows that its u^k coefficient has q -degree at most $\lfloor k/2 \rfloor$, apart from the explicit terms $1 - 2u$. In particular for $k \geq 1$, $[u^k](\delta - 1)$ has q -degree at most $k - 1$. Therefore $z(\delta - 1) \in \mathcal{R}$, with zero u -constant term. Both exponentials in (30.2) belong to $\mathcal{R}$. Also δ , $1/L_1$ and $1/(\delta - s)$ belong to $\mathcal{R}$. The series $\delta + s$ starts at u^2 , and its u^k coefficient has q -degree at most $k - 1$. Thus both $q(\delta + s)/(\delta - s)$ and $4uq/(\delta - s)$ belong to $\mathcal{R}$. So do $A, T_1, (1 + A)^{-1}$ and P . It follows that

$$[u^k]P(z, u) = \sum_{j=0}^k P_{k,j}(z) e^{jz}, \quad \deg P_{k,j} \leq k - j.$$

Here $q = e^z$ is substituted only after polynomial coefficient extraction. At formal $q = 0$, $A = T_1 = 0$ and $\delta = 1 - 2u$, so

$$P|_{q=0} = \frac{1 - 2u}{1 - u} = 1 - \sum_{k \geq 1} u^k.$$

Consequently, for every $k \geq 1$,

$$[u^k]P(z, u) = -1 + \sum_{j=1}^k P_{k,j}(z)e^{jz}, \quad \deg P_{k,j} \leq k - j. \quad (30.3)$$

The constant -1 is consistent with $[u^k]P(0, u) = 0$, because the other terms cancel it at zero.

For $j \geq 1$ and $\ell \geq 0$,

$$n![z^n](z^\ell e^{jz}) = (n)_\ell j^{n-\ell}, \quad \sum_{n \geq 0} (n)_\ell j^{n-\ell} x^n = \frac{\ell! x^\ell}{(1 - jx)^{\ell+1}}.$$

The summand is zero for $n < \ell$. Since $\ell \leq k - j$, each denominator divides D_k . Each rational summand is proper, so $F_k(x) = -1 + \text{proper rational function}$. Multiplying by D_k , all terms other than $-D_k$ have degree below $\deg D_k = k(k+1)/2$. The leading coefficient of $-D_k$ is the stated sign times $\prod_{j=1}^k j^{k-j+1} = \prod_{j=1}^k j!$. Finally $D_k F_k$ has integer formal coefficients and is already known to terminate, so $N_k \in \mathbb{Z}[x]$. $\square$

This proof does not determine which individual poles survive cancellation or the minimal denominator. It also supplies no new complexity bound for computing the full refined triangle.

31 Exact checks and source provenance

The first complete polynomials are

$$\begin{aligned} p_0(u) &= 1, & p_1(u) &= p_2(u) = u, \\ p_3(u) &= u + 2u^2, & p_4(u) &= u + 8u^2 + 2u^3, \\ p_5(u) &= u + 22u^2 + 26u^3, \\ p_6(u) &= u + 52u^2 + 168u^3 + 42u^4, \\ p_7(u) &= u + 114u^2 + 804u^3 + 692u^4 + 42u^5. \end{aligned}$$

These rows and fixed-run rational formulas are prior data, not novelty claims. The companion expands (21.3) directly in $\mathbb{Q}[u][[z]]$ using exact fractions and exact polynomial division by the initial denominator $4u - 1$. Multiplication by $n!$ gives integer polynomials. This independent expansion is compared against the published run-refined endpoint recurrence [3, Eq. (3)] through $n = 40$, coefficient by coefficient, rather than by interpolation at a few values of u .

The second check literally reverses the maximal decreasing runs of every input permutation through $n = 9$, deduplicates all resulting outputs, and counts their maximal ascending runs. This route does not use the overlap characterization. A third check multiplies the published fixed-run ordinary denominators for $k = 1, \dots, 5$ against the computed columns [2, pp. 10–11]. Finally optional formal checks of (30.2) through $k = 12$ verify the degree filtration, constant q term, exponential-polynomial counts, integer numerator degree and leading coefficient. The slope certificate uses only the rational arithmetic in Section 28.

The source package distinguishes published fixed- k fixtures from computed outputs and records exact source URLs, locations and snapshot hashes. Public research snapshots of full papers and of Patel’s post were used for source checking; they are not repackaged as authored report content. The complete claim text was read, not merely the MathDB generated progress summary. The latter’s model label is not a proof-author attribution.

31.1 Bounded literature findings

The overlap characterization originates in [1], with the later development [2]. Theorem 6 of [2] proves fixed-run rationality; Theorem 10 is a five-variable ordinary functional equation, including run marking, and Theorem 11 gives enumeration recurrences. These are distinct from the explicit bivariate EGF here. The run-refined endpoint recurrence and fixed- k denominator conjecture are in [3]; its numerical total-count asymptotics are also prior art. No improvement over its stated refined-count operation count is claimed.

The published 2023 version of *Semidistrim Lattices* [5], Section 11.2 and Figure 12, gives initial type-A image polynomials with the ascent statistic. For permutations of $[n]$, those polynomials are $u^n p_n(1/u)$, because the ascent count is $n - K_n$; the reversal matters when comparing rows. The OEIS triangle A309993 [10] supplies initial rows and the first five fixed-run functions. Its linked 5050-term b-file was not retrieved in the source audit and is not claimed checked.

[write] **6 October 2026.** The write read A309993 (revision #21 of August 29, 2019, by Bjarki Ágúst Guðmundsson): its column generating functions for $k \leq 5$ have the numerators $N_1, \dots, N_5$ of [2, p. 10] used as fixtures here (compared coefficient by coefficient), and its b-file covers $n \leq 100$. The b-file was not checked by the write either. Asinowski–Banderier–Hackl’s Theorem 6 (p. 8) and the formulas (p. 10) are where this Part cites them.

A bounded search on 3 October 2026 checked those primary sources and queries combining pop-stacked permutations, runs, bivariate EGF, exponential generating function, variance, central limit, local limit, and denominator conjecture. No equivalent explicit bivariate EGF or run-count Gaussian law was located. This supports only the statement that those results go beyond the sources checked, not an exhaustive worldwide novelty or priority certification. Patel’s earlier denominator proof is explicitly credited in Section 30. Broader scholarly review remains appropriate before publication. No publication, author contact, OEIS edit, or repository modification was performed.

The spectral, analytic-perturbation, quasi-powers and Fourier techniques are standard. References [6, 8, 7] identify this background; the problem-specific normalization, cancellation, complex uniformity, variance calculation and lattice estimates are proved above.

32 Companion and reproducible package

The companion uses Python’s standard library only. Its mathematical checks remain active under `python -O`. Public computation commands have bounded inputs, produce deterministic JSON on standard output, and perform no file or network I/O. Shell redirection is the caller’s optional choice for saving a result. Separate release helpers handle the explicitly requested local PDF and ZIP outputs.

The package includes the editable L^AT_EX report source, the PDF, companion and tests, published numerical fixtures, computed certificates, source provenance and reproduction scripts. Full-polynomial checking, literal-image checking, published-column checking and optional fixed-run filtration have separate reported scopes. None is a claim of a general formal proof or of numerical effectiveness for (21.8)–(21.10).

The PDF builder uses a fresh exclusive output directory, fixed metadata, disabled TeX shell escape, a bounded reference-stabilization loop and settled log checks. The ZIP builder verifies a fixed SHA-256 allowlist and uses fixed member metadata and exclusive no-clobber output. The local I/O helpers reject symbolic-link path components and pin opened parent directories. They assume

trusted source files and output parents; they are not a sandbox for hostile TeX or an adversary able to replace directory contents. Normal and optimized tests, two independent PDF builds, two ZIP replays, and all-page rendered inspection form the release checks. Temporary files, logs and rendered page images are excluded from the source ZIP. Exact commands and receipts accompany the package.

[write] In this directory the companion is `code/166-runs-companion-*.py` with its README and provenance notes (`166-runs-companion-README.md`, `166-runs-companion-PROVENANCE.md`), the data are `data/166-runs-data-*.json`, and the release tools are `code/166-runs-*.py`. The delivered README, `Report166.tex`, the PDF and SHA256SUMS are not shipped. Nothing runs in this directory under the shipped names; the report's README gives the rerun route.

33 Further questions and research

This section is added in the write, under Vladimir’s standing rule of 4 October 2026: every claim of the three manuscripts that is not proved here, and every limitation they state, is collected as a question, with its source, the sketch that came with it, and what is missing. Questions that a later Part answers are recorded as answered, not dropped.

1. *Effective complex gap, error constants and onsets.* Parts I and II prove $p_n = C n! \rho^{-n} (1 + O(\theta^n))$ for some $\theta < 1$ (Part I, Theorem 9.2; Part II, Theorem 14.1); Part III proves its moment, central and local limit estimates with unspecified constants and onsets; all three say that no numerical value of R , of the error constant or of the onset is certified, and that the real-interval certificates do not supply one. *Sketch:* the pole set is explicit, $\{D = 0, Q \neq 0\}$, so a certified argument-principle count of the zeros of the entire function D in a disk $|z| < R$ would give an explicit R , and Cauchy’s estimate on $|z| = R$ an explicit constant. Numerically, the second pole by modulus is $\rho_1 = 2.417\dots$: a floating-point argument-principle count by the write (30 digits, not certified) finds exactly one, two and three zeros of D in $|z| < 2, 3$ and 3.5 (these are ρ_0, ρ_1, ρ_2 ; the only zero of Q in these disks is $-\log 4$, where $D = 2$; the others, $-\log 4 + 2\pi i k$, have modulus above 6.4), and the nonreal poles known have moduli at least $|0.4279 + 3.6013i| = 3.627$. *Missing:* the certified count and a certified bound for $|P|$ on the circle; for Part III, uniformity of these in u .
2. *A finite-input threshold certificate* (Part II, Section 20; Part I, Section 11). Proposition 11.1 and Theorem 19.2 are exact implications whose constants M, θ, n_0 are not effective. *Missing:* item 1, after which the bracket becomes executable for every y beyond an explicit bound, with exact counting below it.
3. *The nonreal poles* (Part I, Section 6). Part I gives the equation $D = 0$, the simplicity and the residues, but no ordering of the nonreal poles by modulus and no convergent expansion over all poles; (6.3) is finite-radius only. Claesson–Guðmundsson–Pantone’s numerics and the authors’ repository list eight nonreal pairs with real parts increasing from 0.43 to 4.13 and imaginary parts from 3.60 to 4.61 (Remark 5.1). *Missing:* a description of the nonreal zeros of D (location, asymptotic distribution, ordering) and bounds for P between them that would justify or refute a convergent pole sum.
4. *A minimal differential equation* (Part I, Section 7). The resultant gives an algebraic differential equation of order at most two; Part I asserts neither minimality nor that it determines P among its solution germs. *Missing:* whether order one is impossible (that is, whether P satisfies no algebraic first-order equation over $\mathbb{Q}(z)$), and a minimal polynomial of the given order.
5. *Bit complexity and optimal cost* (Part I, Section 8; Part III, Section 31). Part I’s recurrence uses $O(N^2)$ arithmetic operations on integers of $O(N \log N)$ bits; it claims no bit-complexity bound, no wall-clock guarantee and no optimality, and Part III claims no improvement of the run-refined count. Asinowski–Banderier–Hackl’s conclusion (p. 37) asks for the optimal cost of computing pop-stacked permutations. *Re-scoped:* the arithmetic operation count for the total count is lowered from $O(N^4)$ to $O(N^2)$; the optimal cost, a bit-complexity bound, and a faster run-refined algorithm (for instance from Part III’s bivariate formula or a bivariate Riccati equation) remain open.

6. *The bivariate polar set* (Part III, Sections 21 and 24). Part III proves the formula as a single-valued meromorphic continuation but does not classify the zeros of $D(z, u)$ that are poles, nor the exceptional parameters such as $u = 1/4$ where numerator and denominator vanish together. At $u = 1$ Part I answers this. *Missing*: the analogue of Part I's pole-set theorem for general u (a Riccati equation in z with parameter u would be the natural route).
7. *Minimal fixed-run denominators* (Part III, Section 30). Theorem 30.1 and Patel's proof give a denominator that works for every k , not the reduced one, and do not say which poles survive cancellation. Claesson–Guðmundsson–Pantone found rational fits for $k \leq 24$ (Section 3.2); Patel's value $N_k(1) \neq 0$ shows that the factor $(1-x)^k$ survives in full. *Missing*: the multiplicity of each factor $1-jx$ in the reduced denominator for general k .
8. *Finer limit laws* (Part III, Sections 21 and 29). Part III proves the central limit theorem and a lattice local limit theorem with absolute error $O(n^{-1})$; it claims no Edgeworth expansion, no global large deviations, no uniform relative error in the tails and no certified onset. *Sketch*: the quasi-power form (27.2) holds uniformly for complex u near 1, which is the usual starting point for Edgeworth corrections and for large deviations in a neighbourhood of the mean. *Missing*: the higher cumulant terms with uniform error, and control of $\rho(u)$ for u away from 1.
9. *Certified β and γ* (Part III, Section 27). Part III defines the constant terms exactly and evaluates them numerically without a certificate; the write's independent evaluation agrees with all printed digits (note in Section 27). *Missing*: rational interval enclosures, by the same method as Part III's certificate for μ and v applied to $C'(1)$ and $C''(1)$.
10. *The image of a uniformly random input* (Part III, Section 21). Part III studies the run count of a uniformly chosen distinct output and says that weighting outputs by their numbers of preimages gives a different distribution, not considered. *Missing*: any statement about that distribution.
11. *Data not checked*. The 5050-term b-file of A309993 was checked neither by Part III nor by the write. The 450-term b-file of A307030 was checked by the write against Part I's recurrence (no mismatch); the journal version of Claesson–Guðmundsson–Pantone was not read.
12. *Literature and priority*. All three Parts made bounded source searches on 3 October 2026 and make no worldwide priority claim; Part I says that scholarly review is appropriate before any submission. The write found one omission of credit, recorded in place: Part II's interleaving bound is Asinowski–Banderier–Hackl's Theorem 12. For Conjecture 2, Patel's proof has priority over Part III (Section 30). *Missing*: a search beyond the sources the three Parts and the write read.

[write] **6 October 2026.** *Proved, refuted and corrected in this report.* The predictions of Claesson, Guðmundsson and Pantone (simple poles, infinitely many singularities, non-D-finiteness of both generating functions, the constants ρ and C , the equivalent $p_n \sim Cn!\rho^{-n}$) are proved by Part I, and Conjecture 2 by Patel and, by a different proof, Part III. Their printed list of predicted positive poles omits $\rho_{12} = 5.33887671271979\dots$ (Remark 5.1, interval proof; their repository lists it), and their three printed nonreal pairs are correct to decimal places 62–66 and then lose two digits each (Remark 5.1, floating-point computation). Part II's statements that further poles, non-D-finiteness and certified decimals are not proved are superseded by Part I (dated notes, sentences kept). No mathematical statement of the three manuscripts was found wrong.

[write] **Independent check of the write, 6 October 2026.** An adversarial check made by the intake after the write (commit 12013349a) re-derived the statements of the write from its sources. *Remark 5.1.*

With its own identification step ($D = R \cos(zr/4 + \alpha)$ for $z > 0$, $\alpha \in (\pi/6, \pi/4)$) and `mpmath.iv` at 130 digits, it found the same fifteen sign changes and the same indices $k = 0, \dots, 11, 13, 14, 15$, and $k = 12$ at 5.3388767127; the table of $\rho_0, \dots, \rho_{19}$ matches 230-digit roots, and the printed decimals of C and $1/\rho$ are correct truncations. The repository file [12] has exactly one commit, 9c58ae6 of 20 August 2019, and is unchanged since; its twenty positive values lie within their stated last-digit uncertainty, $\rho_4, \dots, \rho_{19}$ by 230-digit roots and $\rho_0, \dots, \rho_3$ (252 to 782 decimals) by 900-digit roots. Newton’s method at 260–300 digits on the branch-free form of D reproduced the three nonreal pairs, their first wrong printed decimals and the two deleted digits of each string. *Patel*. The account, the page metadata and every step of the argument were confirmed; an independent word-model computation of $N_k = D_k F_k$ reproduces the published numerators for $k \leq 5$, and the leading coefficients and $N_k(1)$ for $k \leq 6$. *Other*. The credit of Part II’s interleaving bound to Asinowski–Banderier–Hackl’s Theorem 12 (p. 18) and the value $(24/23)^{1/4}$ were confirmed, and so were the four classifications against the transseries volume. Part I’s recurrence (8.1), in independent code, reproduces all 450 terms of the A307030 b-file, the pop-stack image of all permutations for $n \leq 9$, and the Taylor coefficients of N/D for $n \leq 25$. No mathematical error was found. Four wordings were corrected: (i) ρ_{12} was printed in four places as a rounded value followed by an ellipsis (“5.33887671272...”, “5.338876712720...”), which in this report means a truncation; it is now the truncation 5.33887671271979... (the table of Remark 5.1, which says “rounded”, is unchanged); (ii) the repository’s complex values were said to agree with the zeros “to 110 or more digits”; the real part of the third pair agrees only to 108 decimal places ($|\text{difference}| = 1.8 \cdot 10^{-109}$), so both places now say “at least 108 decimal places”; (iii) the front matter put in quotation marks MathDB’s “own label” “unverified claimed solution”, which is not on the page: the page gives the problem the status “Claimed solved” (“CLAIMED SOLVED - unverified” in the page data), a status of the problem, not a label of the solution, and Part III’s wording, which is Report 166’s, is kept as delivered; (iv) the write’s note on the location of Asinowski–Banderier–Hackl’s costs overcorrected Part I’s citation of their Theorem 11, whose statement includes the $\sim n^4/8$ time and $\sim n^3/3$ memory costs (dated note in Section 8, keeping both wordings). The sentence on the repository’s twenty positive values now names the computation that confirms them.

References

- [1] A. Asinowski, C. Banderier, S. Billey, B. Hackl and S. Linusson, *Pop-stack sorting and its image: Permutations with overlapping runs*, Acta Math. Univ. Comenianae 88(3) (2019), 395–402. [Publisher PDF](#) (Parts I and III); preprint <https://lipn.fr/~cb/Papers/popstack.pdf> (Part II). Theorem 1 and Sections 3–4.
- [2] A. Asinowski, C. Banderier and B. Hackl, *Flip-sort and combinatorial aspects of pop-stack sorting*, Discrete Math. Theor. Comput. Sci. 22:2 (2021). [DOI](#); [publisher PDF](#). Theorems 3, 6, 10 and 11, Section 2.4 and conclusion (Part I); Theorems 3, 6, 10 and 11, fixed-run formulas on pp. 10–11 (Part III).
- [3] A. Claesson, B. Á. Guðmundsson and J. Pantone, *Counting pop-stacked permutations in polynomial time*, Experimental Mathematics, published online 2021, [journal DOI](#) (Part II’s link text: doi:10.1080/10586458.2021.1926001). Preprint [arXiv:1908.08910](#) (2019). [Author PDF](#); page references use this 12-page PDF. Part I: pp. 2–6 and Sections 3.1–3.3; Part III: Eq. (3), Section 3.2 and Conjecture 2.
- [4] Y. Choi and N. Sun, *The image of the pop operator on various lattices*, [arXiv:2209.13695](#). Scope checked against the primary abstract and the published reference in [5] (Part I).
- [5] C. Defant and N. Williams, *Semidistrim Lattices*, Forum Math. Sigma 11 (2023), e50. [DOI](#). Published Section 11.2, pp. 29–30; Figure 12, p. 31; Remark 11.3, p. 33 (Parts I and III). Preprint: [arXiv:2111.08122](#); relevant enumeration section and A307030 table inspected (Part II).

- [6] R. Ehrenborg, S. Kitaev and P. Perry, *A spectral approach to consecutive pattern-avoiding permutations*, J. Combin. 2(3) (2011), 305–353. DOI (Part II); [arXiv:1009.2119](https://arxiv.org/abs/1009.2119); https://www.ms.uky.edu/~jrge/Papers/Ehrenborg_Kitaev_Perry_A_spectral_approach.pdf. Background on spectral integral-operator enumeration; the specific mixed-run construction and calculations are proved in the Parts.
- [7] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. [Author-hosted text](#); Chapter IX, singularity perturbation and Gaussian laws.
- [8] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Classics in Mathematics, Springer, 1995 reprint. DOI. Chapter VII, analytic perturbation theory.
- [9] The OEIS Foundation, *A307030: Number of permutations of $[n]$ with overlapping adjacent runs*, <https://oeis.org/A307030>; [official source entry](#). Revision 63, 18 November 2021; checked 3 October 2026 by Parts I and II, and again at the write (6 October 2026).
- [10] The OEIS Foundation, [A309993](#), run-refined pop-stacked permutation triangle and fixed-run formulas. Checked 3 October 2026 (Part III); revision 21, 29 August 2019, read at the write. The linked long b-file was not checked.
- [11] S. Patel, *The Denominator Conjecture for Pop-Stacked Permutation Generating Functions*, public MathDB solution posted 20 August 2026. [Solution permalink](#). Posted 14:02:52 UTC, edited 14:04:08 UTC; read 3 October 2026 (Part III) and 6 October 2026 (the write). Site status: unverified claimed solution. See Section 30 for the separate mathematical assessment.
- [12] B. Á. Guðmundsson and collaborators, [Pop-stacked permutations repository](#), README and [singularity estimates](#) (added in commit 9c58ae6 of 20 August 2019, the only commit touching that file). Checked 3 October 2026 (Part I) and 6 October 2026 (the write); numerical estimates, not certified bounds.
- [13] ProveIt, *Transseries and inversion*, the repository’s transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; Part 0: `p0:prop:factorial-core`, `p0:prop:operator-form`, `p0:thm:core-reversion`, `p0:thm:staircase`. Added in the write.